



Genesis of the Bayan qagan dongshan Sn-Pb-Zn-Ag deposit, NE China, based on sphalerite chemistry and sulfur isotope

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ABSTRACT

The Bayan qagan dongshan Sn-Pb-Zn-Ag deposit (223,383 t Sn @ 0.755 %; 9,446 t Ag @ 187.46 g/t; 752,944 t Zn @ 1.60 %; 150,828 t Pb @ 0.521 %) is mainly hosted by the contact zones between early Cretaceous granite porphyry and Permian epimetamorphic rock in the southern part of the Great Khingan Range of northeastern China. Nevertheless, the migration and precipitation form of silver in the deposit is undefined. In this study, seven subtypes of sphalerite (SpI, SpII-1, SpII-2a, SpII-2b, SpII-2c, SpII-3a and SpII-3b) were identified according to mineral assemblages in the quartz-tourmaline-sericite-sphalerite stage (I) and cassiterite-sulfide stage (II) of the deposit. SpI is distributed in the granite porphyry as disseminated ores, accompanied by tourmalinization and fluoritization. SpII-1 is predominantly associated with cassiterite, whereas SpII-2 (including SpII-2a, SpII-2b and SpII-2c) mainly coexisted with stannite, in which chalcopyrite exsolution is found. SpII-3 (including SpII-3a and SpII-3b) is mainly characterized by its irregular shape, and intimately associated with galena and Sb-bearing mineral. Representative time-resolved depth profile and inter-element correlation analyses based on the LA-ICP-MS method illustrated that: 1) Fe, Cd, Mn, Co, Cu, In, Ag and Sb are present in solid solutions formed through direct cation substitution as $\text{Zn}^{2+} \leftrightarrow (\text{Fe}^{2+}, \text{Cd}^{2+}, \text{Mn}^{2+}, \text{Co}^{2+})$ and coupled substitution as $\text{Zn}^{2+} \leftrightarrow \text{Cu}^{+} + \text{In}^{3+}$ and $\text{Zn}^{2+} \leftrightarrow \text{Ag}^{+} + \text{Sb}^{3+}$; 2) Pb, Bi, Sn and Te may be intimately associated with galena and Sn-bearing telluride inclusions hosted by sphalerite. The characteristics of trace and minor elements, along with the results of sulfur isotope studies, suggest that: 1) the ore forming temperature of Bayan qagan dongshan deposit gradually decreases and the silver mineralization mainly occurred in the late stage; 2) the changes of physicochemical conditions (e.g., temperature and pressure) are likely to be an important depositional mechanism for Ag in the Bayan qagan dongshan deposit; 3) the Bayan qagan dongshan deposit may belong to magmatic-hydrothermal origin.

1. Introduction

The southern part of the Great Khingan Range (hereinafter referred to as SGKR), in the east of Central Asian metallogenic domain (hereinafter referred to as CAMD)(Fig. 1a, b), is an important Pb-Zn-Ag-Cu-Fe-Sn-REE polymetallic metallogenic province, which comprises substantial well-known deposits and occurrences of tin or silver such as Bayan qagan dongshan giant-sized Sn-Ag-Pb-Zn-Cu-Sb deposit, Shuangjian-zhan giant-sized Ag-Pb-Zn-Cu deposit, Weilasituo large-sized Sn-Zn-W-Cu-Mo-Rb-Nb-Ta-Li deposit, Huanggang giant-sized Sn-Fe-W deposit, Dajing large-sized Sn-Zn-Cu-Pb-Ag deposit, Bianjiadayuan medium-sized Pb-Zn-Ag-Sn deposit, Maodeng medium-sized Sn-Zn-Cu deposit,

Anle medium-sized Sn-Cu deposit and Daolundaba medium-sized Cu-Sn-W-Ag deposit (Fig. 1c). These deposits have been formed in association with late Jurassic to early Cretaceous magmatism. Due to the identical magmatic sources of Cretaceous volcanic rocks and mineralization-associated granitic rocks, Mao et al. (2018) have suggested that the Sn or Ag polymetallic deposits in the SGKR can be correlated to Sn-Ag ore system related to the subvolcanic rocks, which are similar to those in the Bolivia Sn ore belt. Recently, the genesis of these Sn or Ag polymetallic deposits has been studied (Wang et al., 2016; Yao et al., 2017; Zhai et al., 2017; Chen et al., 2021; Wu et al., 2021), but those studies mainly focused on the geological characteristics, metallogenic epoch and ages of magmatic rocks. Detailed documentation of silver ore-formation

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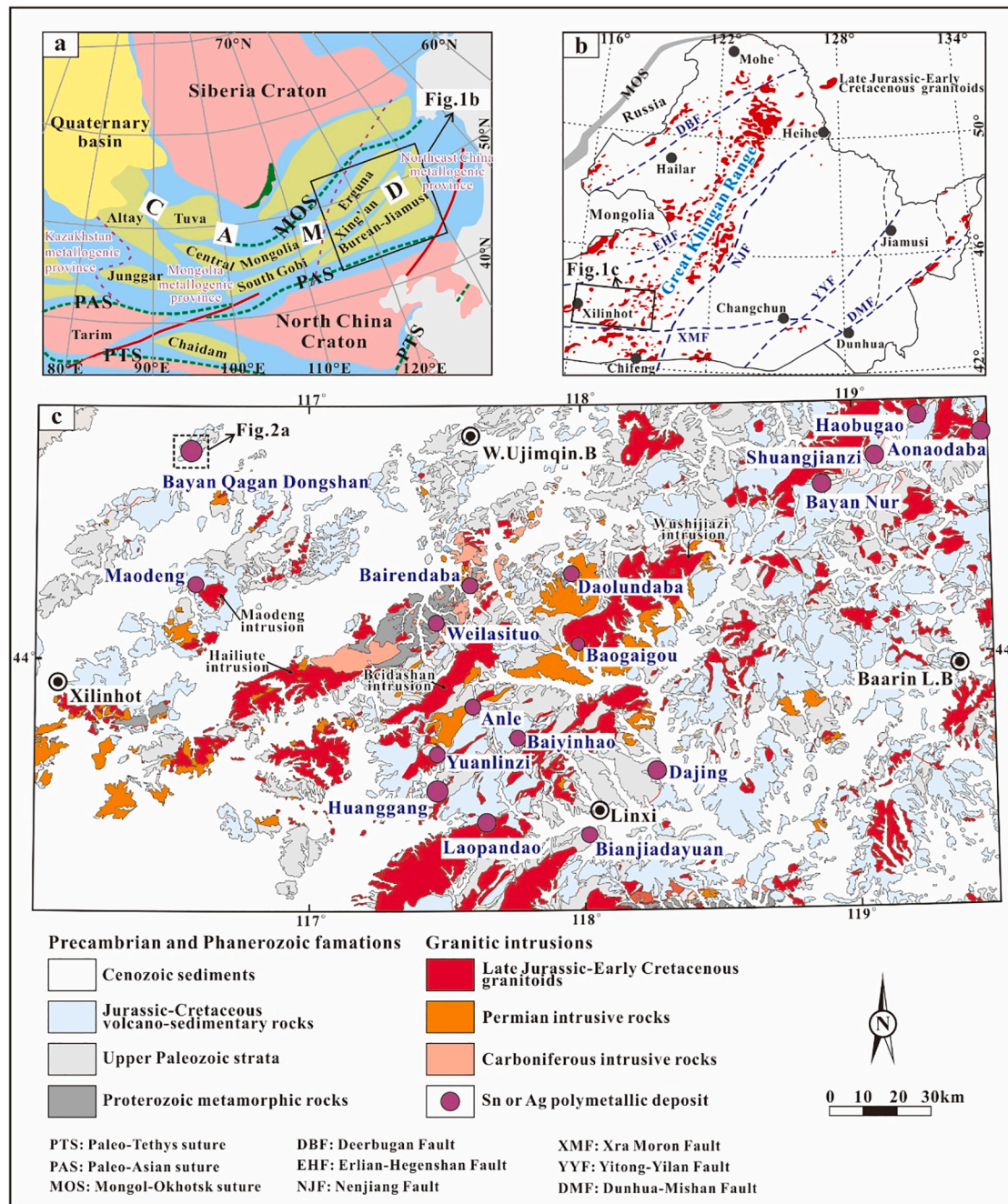


Fig. 1. (a) Tectonic subdivision of the NE Asia (modified after Li et al., 2021a), (b) Sketch map showing the structure and late Jurassic to early Cretaceous granite in the NE Asia (modified after Li et al., 2021a), and (c) Geological map of the southern Great Khingan Range, showing the distribution of Sn polymetallic deposits (modified after IMGSI, 2019).

processes, especially those intimately associated with tin deposit, nevertheless remain scarce. Traditionally, it is not very common for tin and silver to reach large or even super large-scale deposits at the same time, and only Cerro Rico, Deputatskoe and Bayan qagan dongshan were reported at present (Li et al., 2019). The Cerro Rico and Deputatskoe, located in the Bolivia Sn-Ag ore belt, have been well studied. However, the formation process of silver in the Bayan qagan dongshan Sn-Ag polymetallic deposit is undefined, and requires to be furthered studied. Accordingly, we take sphalerite as the research object to reveal the metallogenic mechanism silver in Bayan qagan dongshan Sn-Ag polymetallic deposit, and to provide mineralogical evidence for mineralization study of the Sn-Ag polymetallic ore deposits in the SGKR.

Sphalerite, a zinc sulfide mineral and primary Zn ore mineral, can

incorporate a wide variety of elements in minor and trace concentration within its structure. Typical elements within its crystal lattice comprise Fe, Mn, Co, Cd, Ga, In, Ge, Sn, Cu, Pb, Ag, Bi, As, Se, Hg and Tl, which substitute for Zn cations and S anion (Cook et al., 2009, 2011; Belissant et al., 2014; Ye et al., 2011). The composition and concentrations of these trace elements, as well as isotope composition, especially those document differences mineralization, can reflect sphalerite formation physico-chemical conditions and even the genetic types of deposit (Cook et al., 2009; Xing et al., 2021). In this paper, we analyzed the in-situ geochemical composition and sulfur isotope for different types of sphalerites from Bayan qagan dongshan Sn-Ag polymetallic deposit using Electron Probe Micro Analyzer (EPMA) and Laser Ablation Inductively Coupled Plasma Mass Spectrometry (LA-ICP-MS) methods.

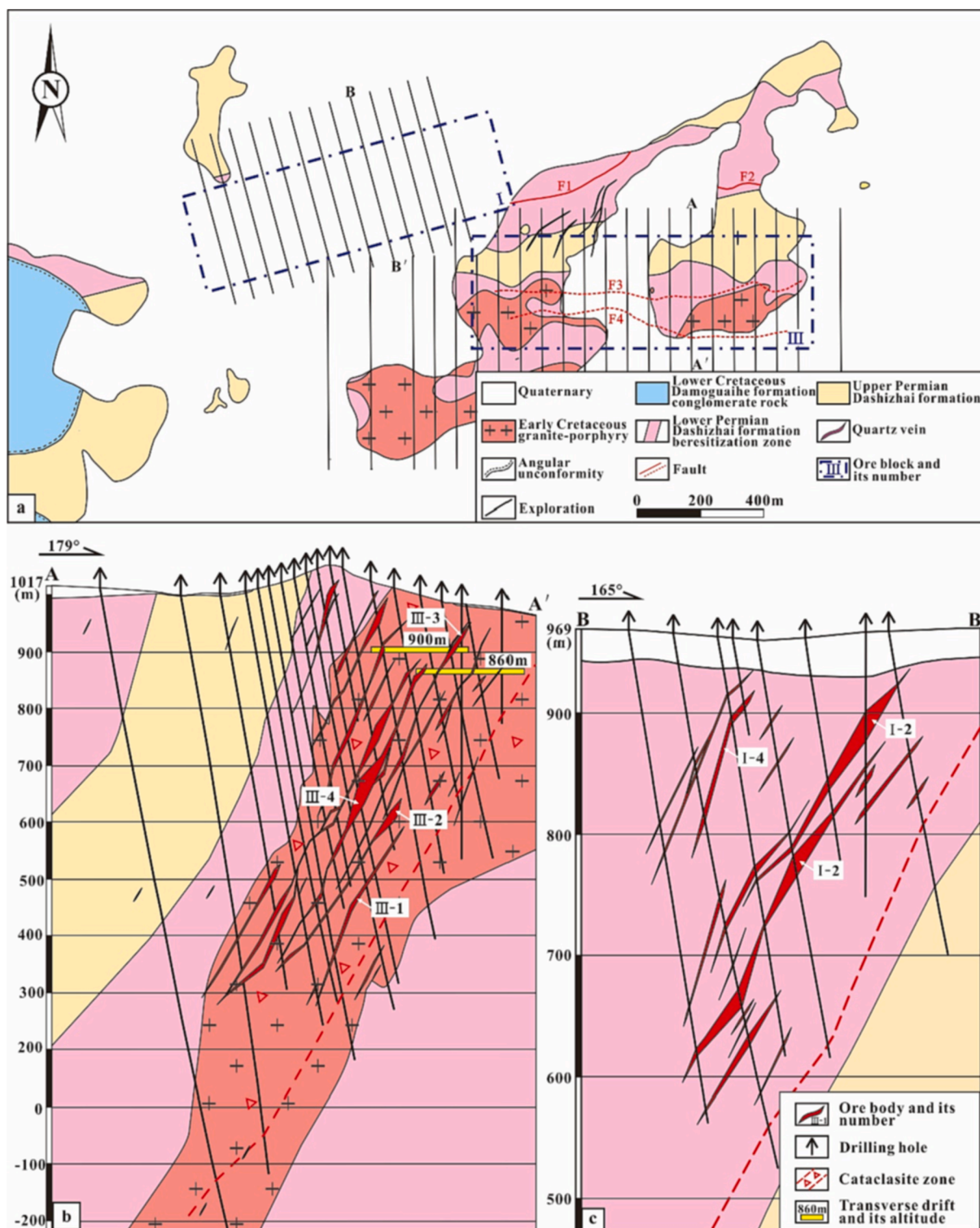


Fig. 2. (a) Simplified geological map, (b) geological section along the A-A' exploration line, and (c) geological section along the B-B' exploration line (modified after SPEIGMR, 2014).

The purpose of this work was to determine the trace and minor elements composition of sphalerite in different mineralization stages, to reveal the occurrence state and the fundamental substitution mechanisms of trace elements within them, to provide the evidence for migration and precipitation mechanism of Ag, and to constrain the genetic type of the Bayan qagan dongshan Sn-Ag polymetallic deposit from the perspective of element geochemistry.

2. Regional geology setting

The CAMD, located between the Siberia, Tarim and North China cratons, is one of the three important metallogenic domains in the world (Gao et al., 2019) (Fig. 1a). It is subdivided into the Kazakhstan metallogenic province in west, the Mongolia metallogenic province in middle and the Northeast China metallogenic province in east (Xiao et al., 2019). The Northeast China metallogenic province experienced

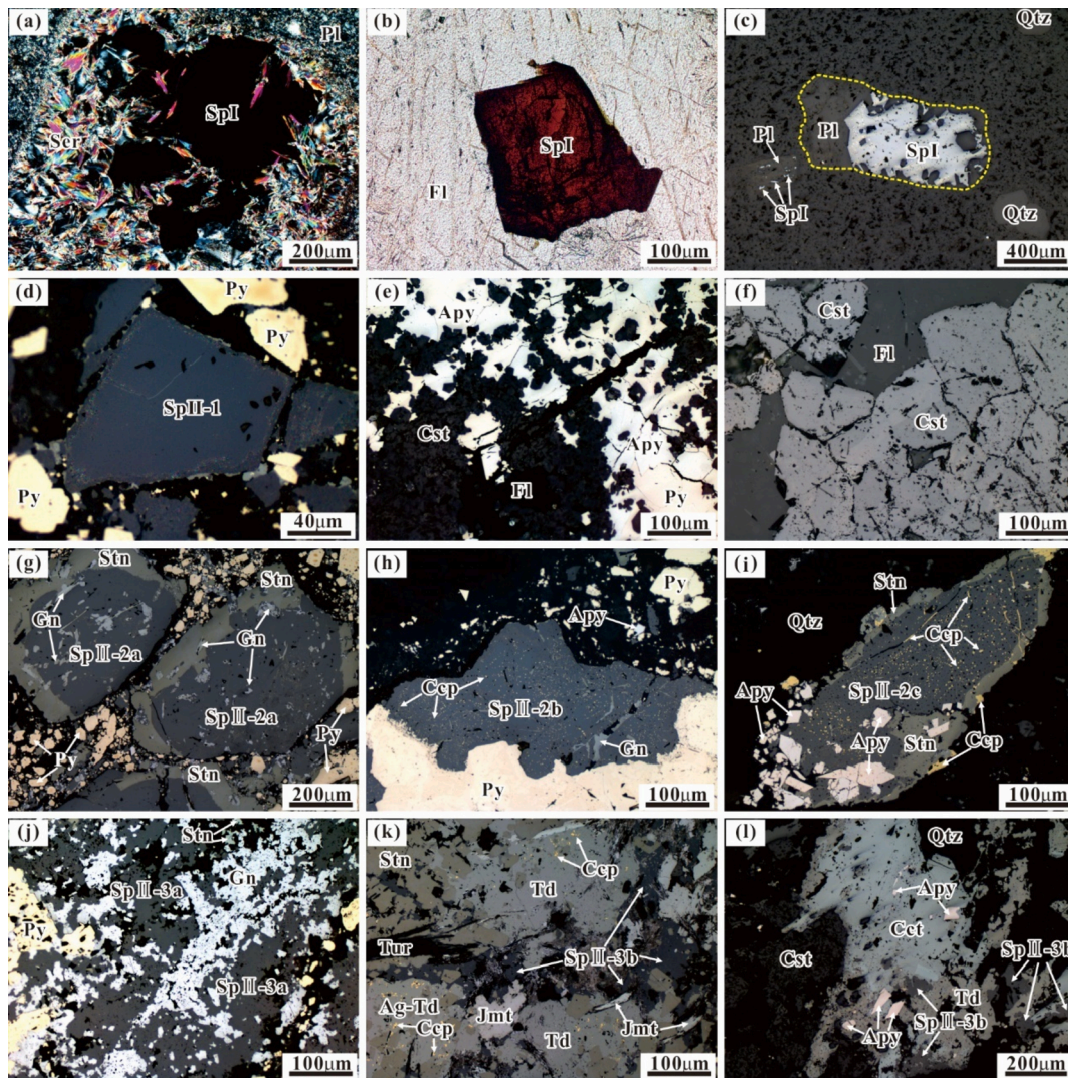


Fig. 3. Photographs and photomicrographs showing mineral assemblages in the Bayan qagan dongshan Sn-Pb-Zn-Ag deposit. (a) Disseminated sphalerite accompanied by Sericite; (b) Disseminated sphalerite accompanied by fluorite; (c) Sphalerite replace plagioclase in granite porphyry; (d) Sphalerite of the stage II-1, accompanied by pyrite; (e) Cassiterite coexists with pyrite and arsenopyrite; (f) Cassiterite coexists with fluorite; (g) The stannite commonly occur along sphalerite II-2a grain boundaries, and sphalerite together with trace amounts of galena; (h) Sphalerite II-2b is crowded with a small amount of micro inclusions of chalcopyrite, and locally crosscut by veinlets containing galena; (i) Sphalerite II-2c is characterized by the development of chalcopyrite inclusions with sizes larger than that of Sphalerite II-2b, and sphalerite II-2c is partially cut by chalcopyrite veinlets and locally contains aggregates of fine-grained arsenopyrite; (j) Sphalerite II-3a commonly appears as irregular intergranular aggregates and coexists with galena and minor amounts of stannite and chalcobite; (k) Sphalerite II-3b intermingled with Sb-bearing minerals such as tetrahedrite, chalcobite, and Jamesonite, occurring as aggregates of fine-grained in general; (l) Sphalerite II-3b coexists with arsenopyrite, chalcopyrite and tetrahedrite. Abbreviations: SpI = Sphalerite I, SpII-1 = Sphalerite II-1, SpII-2a = Sphalerite II-2a, SpII-2b = Sphalerite II-2b, SpII-2c = Sphalerite II-2c, SpII-3a = Sphalerite II-3a, SpII-3b = Sphalerite II-3b, Pl = Plagioclase, Ser = Sericite, Qtz = Quartz, Fl = Fluorite, Tur = Tourmaline, Cst = Cassiterite, Py = Pyrite, Apy = Arsenopyrite, Gn = Galena, Stn = Stannite, Ccp = Chalcopyrite, Td = Tetrahedrite, Ag-Td = Ag-Tetrahedrite, Jmt = Jamesonite, Cct = Chalcobite.

several periods of tectonic reworking during the Paleozoic and Mesozoic, including post-collisional process of the Paleo-Asian Ocean and extension related to the *syn*-subductional tectonics of Mongol-Okhotsk Ocean and *syn*-subductional process of the Paleo-Pacific Ocean (Gao et al., 2019; Chen et al., 2021). The extension event was accompanied by a period of magmatic activity i.e. emplacement of granitic plutons between 135 and 150 Ma (Mao, et al., 2018) (Fig. 1b). These granitic intrusions or granitoids is closely associated with Sn polymetallic mineralization in the Northeast China metallogenic province, especially in the SGKR.

The SGKR occurs in the southwest part of the Northeast China metallogenic province, bounded by Erlian-Hengshan, Xar Moron and the Nenjiang fault (Fig. 1b). Strata exposed in the SGKR mainly comprise of Proterozoic metamorphic basement and Neopaleozoic to Cenozoic

sedimentary covers (Fig. 1c). The Proterozoic metamorphic basement is dominated by gneisses and schists, mainly located in the central segments of the SGKR. The Neopaleozoic cover rocks, consisting of the Shoushangou, Dashizhai, Zhesi and Lixin Formations, are the majority of the wall rocks for many tin and silver polymetallic deposits in the SGKR. The magmas in the SGKR are mainly controlled by NE trending fault, and can be divided into the Carboniferous, Permian, and Late Jurassic-Early Cretaceous periods. The Late Jurassic-Early Cretaceous intrusive rocks are composed of granodiorite, quartz diorite, adamellite, syenite granite and alkali granites; among them, some intrusive rocks are genetically connected to tin and silver polymetallic mineralization in the SGKR. These intrusive rocks are generally characterized by high silicon, rich alkali, extremely negative Eu anomalies and high-fractionated.

In the SGKR, more than 16 Sn dominated deposits and 62 Ag

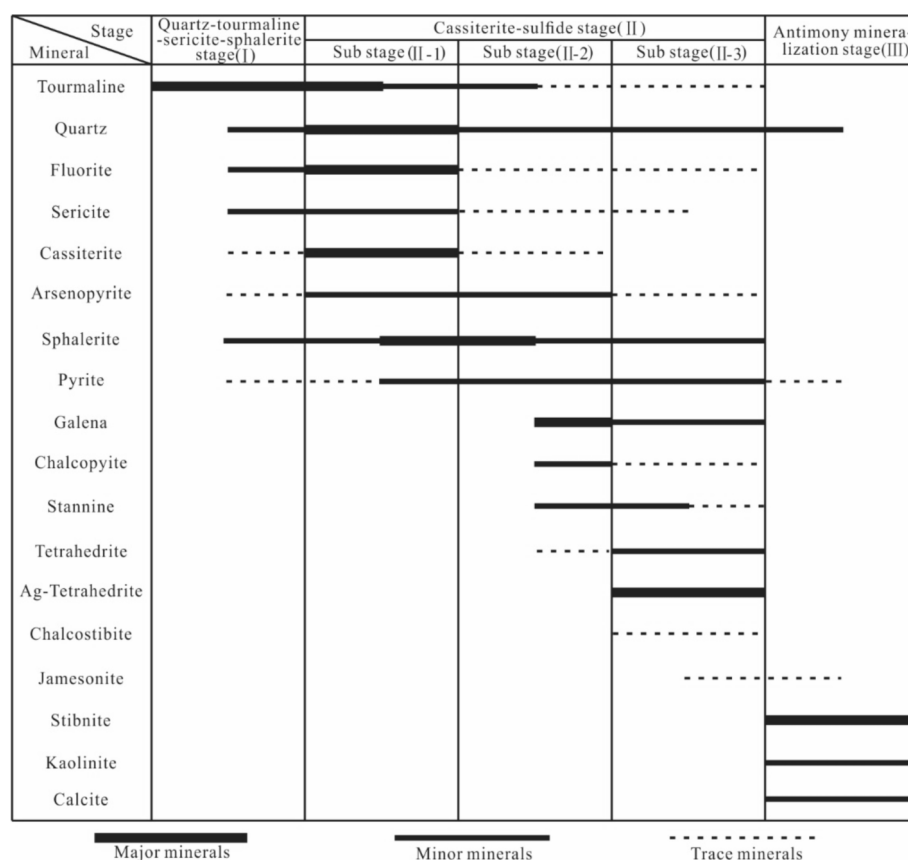


Fig. 4. Paragenesis of the mineral assemblages, including the mineralized sequence of the Bayan qagan dongshan Sn-Pb-Zn-Ag deposit.

dominated deposits are known, including 8 large and super-large deposits. Almost all deposits occur in the Permian strata, with the orebodies controlled by NW, NE or NEE trending fault structure. Spatially, both tin and silver deposits are mostly distributed in the central and western part of the SGKR. This is mainly related to the distribution of Late Jurassic-Early Cretaceous intrusive and sub volcanic rocks. With regard to mineralization age, the tin or silver polymetallic deposits are mostly between 148 and 133 Ma (Wang, 1997; Zhang, 1993; Chen et al., 2021; Wang et al., 2016; Liu, et al., 2018a; Yao et al., 2016; Liao et al., 2014; Wang and He, 2013; Zhai, et al., 2017), broadly contemporary with ages for the Cretaceous magmatism.

3. Deposit geology

3.1. Stratigraphy, structures and magma

The Bayan Qagan Dongshan deposit (lat. 44°32'15", long. 116°33'30") is a giant-sized Sn-Ag-Pb-Zn-Cu-Sb deposit dominated by Sn, Ag, Pb, Zn. It contains proven reserves of ~223,383 tons Sn, ~9,446 tons Ag, ~752,944 tons Zn and ~150,828 tons Pb, with average grades of 0.755 % Sn, 187.46 g/t Ag, 3.22 % Zn, 1.60 % Pb, respectively.

The Bayan Qagan Dongshan deposit is located southwestwards of the SGKR (Fig. 1c). The sedimentary rocks are dominated by the early Permian Dashizhai tuffaceous siltstone, tuff, rhyolite and metamorphic sandstone, mainly occurring in the middle and southwest parts of the ore district. Above the Permian Dashizhai Formation, it is covered with conglomerate in the western part of the mining area (Fig. 2a). The Quaternary sediments are widely distributed and mainly exposed in low and gentle depression of the ore district (Liu, et al., 2017a).

In the southeast part of the ore district, the granite porphyry stock intruded the Permian metamorphic volcanic-sedimentary rocks with small outcrops observed. The granite porphyry intrusions are gray in

color and has a porphyritic texture and massive structure. It mainly comprises quartz, feldspar and biotite phenocrysts and felsic matrix. Most feldspar and biotite phenocrysts have undergone hydrothermal alteration, such as tourmalinization, fluoritization and sericitization. As a result, the original feldspar and biotite only retain the crystal form. Previous studies have proposed that the granite porphyry intrusion yielded ages between 140.2 and 141.7 Ma. It is characterized by high silicon, rich alkali and extremely negative Eu anomalies, and its genetic type belongs to A type granite derived from partial melting of a juvenile lower crust (Liu, et al., 2017b; Yao et al., 2017).

In the ore district, fault with NEE and EW trend developed very well. NEE-trending fault are developed in the Permian tuffaceous siltstone and metamorphic sandstone of the Dashizhai formation (e.g., F1 and F2; Fig. 2a), whereas EW-striking fault, which are mainly concealed, cut granite porphyry close to the contact with the Permian rocks (e.g., F3 and F4; Fig. 2a). The majority of Sn-Ag polymetallic orebodies occur along these EW-striking fault.

3.2. Mineralization

The Bayan qagan dongshan deposit mainly includes two ore blocks, comprising 214 orebodies. Among these, 105 orebodies dominated by Pb, Zn, and Ag are confined by NEE-trending fault in the I ore block and 109 orebodies dominated by Sn, Ag, Cu, and Zn are hosted by EW-trending fault in the III ore block. The ore bodies occur mainly as vein-type in the fractured zones of the Early Cretaceous granite porphyry and the Dashizhai Formation rocks. The NEE-striking No. I-2 and No. I-4, as well as the EW-striking No. III-1 to No. III-4 orebodies are the most important in regard to Sn, Ag, Pb, and Zn reserves. The NEE-striking orebody is represented by the No. I-2. It is 939 m in length along strike, 657 m in depth, 5.70 m in thickness, and dip to the northeast, which contain 1.43 % Pb, 3.17 % Zn and 198.43 g/t Ag

(Fig. 2c). The typical EW-striking orebody is No. III-1, which is 1015 m in length and dips to north with 906 m. The thickness of this orebody is 3.44 m, with an average grade of 0.48 % Sn, 244.12 g/t Ag, 2.77 % Zn and 0.576 % Cu (Fig. 2b).

The ore minerals of the Bayan qagan dongshan deposit primarily comprise cassiterite, sphalerite, galena, chalcocopyrite, stannite and Ag-tetrahedrite, followed by pyrite, arsenopyrite, and stibnite, with minor chalcostibite and jamesonite. The gangue minerals are mainly composed of quartz, tourmaline, fluorite and sericite, followed by calcite, kaolinite, and chlorite. The main ore textures are euhedral-subhedral granular texture, metasomatic dissolution texture, poikilitic texture, cataclastic texture, and exsolution texture. Cassiterite is mainly of euhedral-subhedral granular texture and usually coexists with pyrite, arsenopyrite and sphalerite (Fig. 3e–f). Sphalerite is chiefly produced as subhedral-xenomorphic granular texture and is usually replaced by stannite, developing a metasomatic dissolution texture (Fig. 3e, i). It also contains chalcocopyrite particles, forming an exsolution texture (Fig. 3h, i). Sometimes it is also irregular and coexists with galena and antimony-bearing minerals (Fig. 3j–l). Galena is mainly of xenomorphic granular and irregular vein texture. It usually coexists with or cuts across sphalerite (Fig. 3g–h, j). Chalcocopyrite is primarily of granular and is contained in sphalerite, forming an exsolution texture of sphalerite-chalcocopyrite solid solution (Fig. 3h–i). Stannite is mainly of subhedral granular texture and usually replaces sphalerite. Ag-tetrahedrite is chiefly irregular granular texture and usually replaces stannite, developing a metasomatic dissolution texture (Fig. 3k). It also contains chalcocopyrite particles, forming an emulsion texture (Fig. 3k). Pyrite, arsenopyrite and chalcostibite are mainly produced as euhedral-subhedral granular or irregular shape (Fig. 3e, l). Jamesonite is primarily distributed in stannite in the form of needle column (Fig. 3k). The ore structures are mainly vein structure, fine stockwork structure, disseminated structure, breccia structure, and massive structure; among them, vein and massive structure is the main ore structure.

According to the mineral assemblages, ore structures and crosscutting relationship among the veins, the ore-forming process in the Bayan qagan dongshan deposit can be preliminarily divided into the following paragenetic sequence: quartz-tourmaline-sericite-sphalerite stage (I), cassiterite-sulfide stage (II) and antimony mineralization stage (III). A generalized mineral paragenesis scheme is shown in Fig. 4. Each of the three stages is separately described below.

Stage I is early ore-forming stage in the Bayan qagan dongshan Sn-Pb-Zn-Ag deposit and is characterized by the disseminated sphalerite, tourmaline and a small amount of cassiterite, arsenopyrite and pyrite. The cassiterite commonly appears as irregular intergranular aggregates, coexists with radial tourmaline, and distributes in the granite porphyry as disseminated ores. The sphalerite (SpI) grains are grayish white in color, xenomorphic granular in shape, and range in size from 200 to 600 μm . It usually coexists with tourmaline, fluorite and sericite, which occasionally replace the earlier plagioclase (Fig. 3a–c).

Stage II represents the main mineralization stage and is characterized by three assemblages i.e. quartz-tourmaline-cassiterite-sphalerite-arsenopyrite-fluorite (II-1) (Fig. 3d–f), sphalerite-stannite-chalcocopyrite-galena-arsenopyrite-pyrite-fluorite (II-2) (Fig. 3g–i) and sphalerite-galena-stannite-chalcocopyrite-tetrahedrite-chalcostibite-jamesonite (II-3) (Fig. 3j–l). The sphalerite that is in different assemblages, despite their similar color, differ from each other in a number of characteristics including mineralogy and texture of associated minerals, mineral inclusions, and chemical composition.

Sub stage II-1 is closely associated with the cryptoexplosion and is characterized by the tin-zinc mineralization in the hydrothermal breccia. The breccias generally comprise angular to subangular clasts of their host granite porphyry and silty sand rocks; the matrix is dominated by quartz, tourmaline, fluorite and sericite, along with cassiterite, sphalerite II-1, arsenopyrite and pyrite. Sphalerite II-1 mainly coexists with cassiterite and partially accompanies with pyrite. It mostly occurs as subhedral to anhedral granular aggregates, dark gray-blue in color,

and range in size from 50 to 150 μm (Fig. 3d).

Sub stage II-2 is the main sulfide mineralization stage and is characterized by various sulfides including sphalerite, stannite, chalcocopyrite, galena, pyrite and arsenopyrite, with a medium-high-temperature hydrothermal alteration assemblage such as silicification, fluorite, sericite, and a small amount of tourmalinization. At this stage, sphalerite can be divided into three subtypes, consisting of sphalerite II-2a, sphalerite II-2b and sphalerite II-2c subtypes. Sphalerite II-2a mostly coexists with khaki stannite, several hundreds of microns in size, together with trace amounts of galena. The stannite commonly occur along sphalerite II-2a grain boundaries, indicating that it formed slightly later than sphalerite II-2a (Fig. 3g). Sphalerite II-2b is crowded with a small amount of micro inclusions of chalcocopyrite, and locally crosscut by veinlets containing galena (Fig. 3h). Sphalerite II-2c is typically characterized by the development of chalcocopyrite inclusions with sizes ranging from 2 to 5 μm in diameter, which is larger than that of sphalerite II-2b. In addition, sphalerite II-2c is partially cut by chalcocopyrite veinlets and locally contains aggregates of fine-grained arsenopyrite (Fig. 3i).

Sub stage II-3 is characterized by medium-low-temperature mineral assemblages, consisting of sphalerite, Ag-tetrahedrite, chalcostibite and jamesonite. At this stage, sphalerite is composed of two apparent types dividing into galena-sphalerite (sphalerite II-3a) and tetrahedrite-chalcostibite-jamesonite-sphalerite (sphalerite II-3b) assemblages. Sphalerite II-3a commonly appears as irregular intergranular aggregates and coexists with galena and minor amounts of stannite and chalcostibite (Fig. 3j). However, sphalerite II-3b intermingled with Sb-bearing minerals (e.g., tetrahedrite, chalcostibite, and Jamesonite etc.) generally occurs as aggregates of fine-grained (Fig. 3k–l). A less common, minor amounts of chalcocopyrite inclusions are observed within the slightly brownish gray Ag-tetrahedrite, occasionally occurring in the chalcostibite and stannite.

Stage III is represented by the occurrence of acicular stibnite associating with kaolinite and calcite, which were formed in the latest hydrothermal fluid event. The stibnite is light blue in color, radiated aggregation in shape, and range in size from 200 to 1000 μm . It usually coexists with jamesonite and minor amounts of pyrite, which occasionally has crumple texture.

4. Sampling and analytical methods

4.1. Sampling

Seven representative samples, covering all stages and styles of zinc mineralization, were taken at the 900-m and 860-m level pits of the Bayan qagan dongshan Sn-Pb-Zn-Ag deposit. Each sample was prepared as a polished petrographic thin section, which is used for optical microscope observe, analyzing and distinguishing sphalerite subtypes. Mineral assemblages and sample locations and descriptions are shown in Table A1. In different types of sphalerite, appropriate measurement points of sphalerite were selected for in situ electron microprobe, LA-ICP-MS and sulfur isotopic analyses.

4.2. Analytical methods

Electron microprobe analyses for seven subtypes of sphalerite were obtained using a five-channel JEOL JXA-8230 Electron Probe Micro Analyzer (EPMA) at the MLR Key Laboratory of Metallogeny and Mineral Assessment, Institute of Mineral Resources, Chinese Academy of Geological Sciences, Beijing, China. All measurements were operated at an accelerating voltage of 20 kV, beam current 20nA, and 5- μm beam size. The counting times for peak/background measurement was 10/5. Analytical standards and lines used for analyses were as follows, Zn (ZnS), 184–197 ppm; S (FeS₂), 71–76 ppm; Fe (FeS₂), 176–199 ppm; As (FeAs₂), 232–250 ppm; Se (CuSe), 220–227 ppm; Ag (Au-Ag), 119–173 ppm; Au (Au-Ag), 146–257 ppm; Bi (Bi₂S₃), 406–460 ppm; Cu (CuFe₂S), 160–169 ppm; Ni (NiS), 152–181 ppm; Pb (PbS) 413–421 ppm; Sb

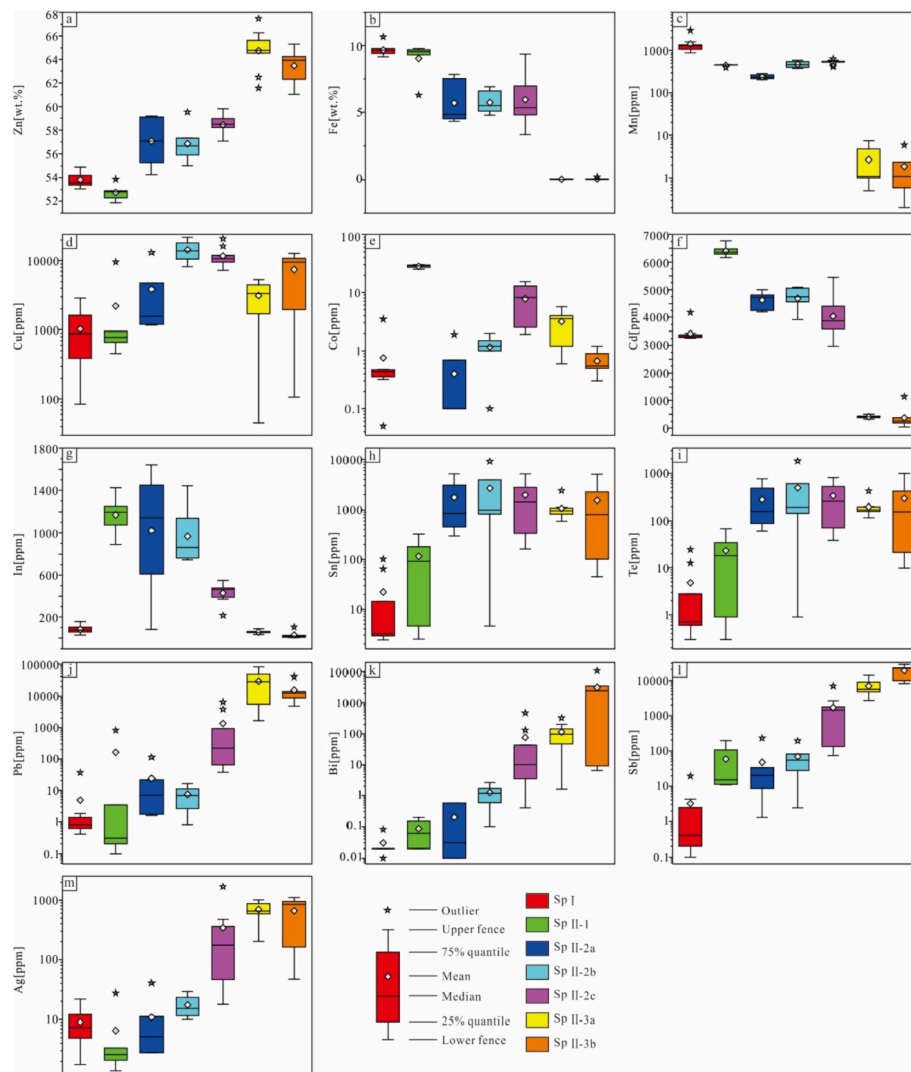


Fig. 5. Box-and-whisker plots of the major and trace element compositions of sphalerite from the Bayan qagan dongshan Sn-Pb-Zn-Ag deposit, grouped into the seven sphalerite subtypes.

(Sb_2S_3), 131–166 ppm; Co (Co), 153–176 ppm.

Laser Ablation Inductively Coupled Plasma Mass Spectrometry analyses of seven subtypes of sphalerite were carried out at the National Research Center for Geoanalysis, Beijing, China. The LA-ICP-MS system consisted of New Wave UP 213 Nd: YAG laser and Finnigan ELEMENT 2 of high-resolution high-sensitivity double-focusing ICP-MS. Helium was used as the carrier gas in the laser ablation system. The laser fluence was 15 J/cm^2 , pulse frequency was 10 Hz, and laser spot beam was $40 \mu\text{m}$. In laser ablation analysis, the first 20 s is a background acquisition, following by 40 s of data acquisition from the sample. The accuracy of elemental analysis, assessed by results of systematic measurements of standard samples NIST612, NIST610, KL2G and MASS-1, did not exceed $\pm 10 \%$, or even $\pm 5 \%$.

The seven subtypes of sphalerite were subjected to in-situ sulfur isotopic analysis at the MLR Key Laboratory of Metallogeny and Mineral Assessment, Institute of Mineral Resources, Chinese Academy of Geological Sciences, Beijing, China. Experimental instruments include a Resolution S155 laser ablation system and a Neptune Plus MC-ICP-MS equipped with 9 Faraday Cup and $10^{11} \Omega\text{m}$ Resistance amplifier. The medium resolution mode is used to remove the influence of interference peaks, and three Faraday Cups including L3, C and H3 are used to receive ^{32}S , ^{33}S and ^{34}S signals at the same time. During the measurements, ablation was performed in pure He gas; laser fluence was 5.0 J/

cm^2 ; frequency was 4 Hz, and spot size was $32 \mu\text{m}$. The NBS123 ($\delta^{34}\text{S}_{\text{V-CDT}} = 17.8 \pm 0.2 \text{ ‰}$) was applied as external standard samples for the calibration of sphalerite, which the accuracy of analysis did not exceed $\pm 5 \%$.

5. Analytical results

5.1. Major elements

The EPMA data of sphalerites in different stages from the Bayan qagan dongshan Sn-Pb-Zn-Ag deposit show values of Zn and Fe concentrations ranges of 51.88–67.48 wt% and 0.02–12.18 wt%, respectively. Sphalerite I has completed 9 EPMA spot analyses, of which the average concentrations of Zn is 53.83 wt% and the average concentrations of Fe is 11.62 wt%; sphalerite II-1 has completed 6 EPMA spot analyses, of which the average concentrations of Zn is 52.75 wt% and the average concentrations of Fe is 11.39 wt%; sphalerite II-2a has completed 7 EPMA spot analyses, of which the average concentrations of Zn is 57.06 wt% and the average concentrations of Fe is 7.71 wt%; sphalerite II-2b has completed 6 EPMA spot analyses, of which the average concentrations of Zn is 56.85 wt% and the average concentrations of Fe is 7.16 wt%; sphalerite II-2c has completed 9 EPMA spot analyses, of which the average concentrations of Zn is 58.46 wt% and

Table 1

Summary of LA-ICP-MS data for the different sphalerite subtypes from the Bayan qagan dongshan Sn-Pb-Zn-Ag deposit (data in ppm).

Sphalerite subtype		⁵⁵ Mn	⁵⁹ Co	⁶⁵ Cu	¹⁰⁷ Ag	⁷¹ Ga	¹¹¹ Cd	¹¹⁵ In	¹¹⁸ Sn	¹²¹ Sb	¹²² Te	²⁰⁸ Pb	²⁰⁹ Bi
Sphalerite I (n = 9)	Mean	1438.2	0.76	1028	9.1	3.2	3421	83	22.2	3.2	4.8	4.8	0.03
	75 % quantile	1383.4	0.48	1618	12	5.1	3383	105	14.4	2.5	2.8	1.4	0.02
	Median	1303.3	0.44	881	7.4	1.7	3328	76	3.2	0.4	0.7	0.8	0.02
	25 % quantile	1064	0.36	387	4.9	1.5	3281	58	2.9	0.2	0.6	0.6	0.02
Sphalerite II-1 (n = 6)	Mean	449.4	28.3	2205	6.6	2.4	6403	1172	116.5	59	23.3	161.2	0.08
	75 % quantile	465	29.8	957	3.4	2.9	6482	1253	180.4	107	34.4	3.4	0.15
	Median	456.7	28.8	784	2.6	2.3	6360	1196	92.5	15	18.0	0.3	0.06
	25 % quantile	454.4	27.0	669	2.1	1.8	6281	1076	4.6	11	0.9	0.2	0.02
Sphalerite II-2a (n = 7)	Mean	240.4	0.4	3846	10.8	12.6	4622	1023	1731	48	275	23.9	0.20
	75 % quantile	266.3	0.7	4736	11.3	14.8	4812	1451	3075	34	486	21.6	0.57
	Median	232.0	0.1	1566	5.2	12.3	4725	1142	849	20	158	6.9	0.03
	25 % quantile	221.4	0.1	1193	2.8	10.5	4245	611	453	9	88	1.7	0.01
Sphalerite II-2b (n = 6)	Mean	475.0	1.2	14,403	17.4	3.5	4687	969	2654	69	489	7.4	1.3
	75 % quantile	546.1	1.5	18,027	23.2	3.2	5053	1139	3923	83	599	11.1	1.7
	Median	462.6	1.2	13,851	15.3	2.8	4741	862	997	55	192	6.7	1.2
	25 % quantile	407.4	1.0	10,553	11.6	2.2	4562	763	817	28	143	2.6	0.6
Sphalerite II-2c (n = 9)	Mean	537.2	7.8	11,701	342.7	8.7	4037	430	1952	1699	332	1336	75.8
	75 % quantile	562.9	12.8	11,931	360.2	9.9	4401	478	2758	1785	516	918	42.9
	Median	543.8	8.1	10,739	174.9	9.1	3870	464	1410	1448	253	222	10.1
	25 % quantile	533.8	2.5	9518	46.5	8.0	3567	388	341	134	70	64	3.5
Sphalerite subtype		⁵⁵ Mn	⁵⁹ Co	⁶⁵ Cu	¹⁰⁷ Ag	⁷¹ Ga	¹¹¹ Cd	¹¹⁵ In	¹¹⁸ Sn	¹²¹ Sb	¹²² Te	²⁰⁸ Pb	²⁰⁹ Bi
Sphalerite II-3a (n=9)	Mean	2.7	3.1	3112	696.4	3.1	415	59	1073	7008	196	29,311	113.1
	75% quantile	4.8	4.0	4440	873.1	3.6	440	67	1076	8975	197	48,200	143.2
	Median	1.1	3.5	3355	656.7	2.7	413	53	967	5602	171	28,031	95.9
	25% quantile	1.0	1.2	1689	588.2	1.9	377	50	821	4724	158	5506	47.4
Sphalerite II-3b (n=6)	Mean	1.9	0.7	7438	662.2	0.8	380	31	1516	18,657	392	15,404	3146.5
	75% quantile	2.4	0.9	10,845	955.9	1.0	383	26	2263	22,237	418	13,998	3541.1
	Median	1.1	0.6	9513	848.1	0.9	258	21	808	21,881	155	12,537	2443.3
	25% quantile	0.6	0.5	1962	163.4	0.5	196	9	103	9948	21	8787	9.0

the average concentrations of Fe is 6.39 wt%; sphalerite II-3a has completed 9 EPMA spot analyses, of which the average concentrations of Zn is 64.76 wt% and the average concentrations of Fe is 0.08 wt%; sphalerite II-3b has completed 6 EPMA spot analyses, of which the average concentrations of Zn is 63.48 wt% and the average concentrations of Fe is 0.10 wt%. From sphalerite I to sphalerite II-3b, the content of Zn is gradually increasing, while the content of Fe is gradually decreasing (Fig. 5a-b). The contents of all analyzed elements using EPMA are listed in Table A2.

5.2. Trace elements

The trace element concentrations of sphalerites were acquired by selecting only stable signal intervals in LA-ICP-MS spectra, consisting of ⁵⁵Mn, ⁵⁷Fe, ⁵⁹Co, ⁶⁵Cu, ⁶⁶Zn, ⁷¹Ga, ⁷²Ge, ¹⁰⁷Ag, ¹¹¹Cd, ¹¹⁵In, ¹¹⁸Sn, ¹²¹Sb, ¹²²Te, ²⁰⁸Pb and ²⁰⁹Bi. The variations of these elements mainly respond to analyses of solid solutions rather than mixed minerals by analyzing mineral volumes free of obvious inclusions. Results of all elements isotopic determinations in sphalerite from the Bayan qagan dongshan Sn-Pb-Zn-Ag deposit are listed in Table 1 and Table A3.

The average concentration of Mn in sphalerite I (1438.2 ppm), sphalerite II-1 (449.4 ppm), sphalerite II-2a (240.4 ppm), sphalerite II-2b (475.0 ppm), sphalerite II-2c (537.2 ppm), sphalerite II-3a (2.7 ppm) and sphalerite II-3b (1.9 ppm) generally show a trend of decreasing values from sphalerite I to sphalerite II-3b (Fig. 5c). The average Cd content for sphalerite I (3421 ppm), sphalerite II-1 (6403 ppm), sphalerite II-2a (4622 ppm), sphalerite II-2b (4687 ppm), sphalerite II-2c (4037 ppm), sphalerite II-3a (415 ppm) and sphalerite II-3b (380 ppm) also show a trend of decreasing values from sphalerite II-1 to sphalerite II-3b in general (Fig. 5f). In conclusion, the change trend of Mn and Cd is similar to that of Fe (Fig. 5b, c, f).

The average concentrations of Cu in sphalerite I, sphalerite II-1, sphalerite II-2a, sphalerite II-2b, sphalerite II-2c, sphalerite II-3a and sphalerite II-3b were 1028, 2205, 3846, 14403, 11701, 3112, and 7438 ppm, respectively. The results show a trend of increasing values from sphalerite I to sphalerite II-2b, then decreasing to sphalerite II-3c in

general (Fig. 5d). In contrast to Cu, the mean In content measured by LA-ICP-MS were 83, 1172, 1023, 969, 430, 59 and 31 ppm, respectively. The average In contents show a similar distributional trend as Cu. However, the maximum value of Cu is in sphalerite II-2b, while the maximum value of In is in sphalerite II-1.

The average Sn content for sphalerite I (22.2 ppm), sphalerite II-1 (116.5 ppm), sphalerite II-2a (1731 ppm), sphalerite II-2b (2654 ppm), sphalerite II-2c (1952 ppm), sphalerite II-3a (1073 ppm) and sphalerite II-3b (1516 ppm) show a trend of increasing values from sphalerite I to sphalerite II-2b, then decreasing to sphalerite II-3b (Fig. 5h). The average contents of Te were 4.8, 23.3, 275, 489, 332, 196 and 392 ppm in sphalerite I, sphalerite II-1, sphalerite II-2a, sphalerite II-2b, sphalerite II-2c, sphalerite II-3a and sphalerite II-3b, respectively. The results are consistent with that of Sn, revealing a trend of increase followed by an decrease (Fig. 5i). In addition, the mean Sn and Te contents ratio is constant at 1/5 in each subtype of sphalerite. Both of them show a significantly positive correlation (Fig. 7h).

The average concentrations of Ag, Sb, Bi, and Pb from sphalerite I to sphalerite II-3b exhibit a similar increasing trend (Fig. 5j-m). In sphalerite II-3a or sphalerite II-3b, the concentrations of Ag, Sb, Bi and Pb reach the maximum value of 696.4 ppm, 18657 ppm, 3146.5 ppm, and 29311 ppm, respectively.

The average contents of Ga and Ge are very low in all subtypes of sphalerite (Ga: av. 5.1 ppm; Ge: av. 0.72 ppm), and the average concentrations of As, Se and Tl were close to or below the minimum detection limits. Nevertheless, the concentrations of individual elements can reach or exceed hundreds of ppm.

5.3. In-situ sulfur isotopic compositions

Table A4 presents the results of the in-situ sulfur isotopic analysis applied to the sphalerite from the Bayan qagan dongshan Sn-Pb-Zn-Ag deposit. The sulfur isotope results reveal a range of $\delta^{34}\text{S}$ values varying from -6.0‰ to -1.1‰ ($n = 21$). The $\delta^{34}\text{S}$ values for Sphalerite I ranged between -6.0‰ and -3.6‰ ($n = 3$), with an average of -5.1‰ . The $\delta^{34}\text{S}$ values for Sphalerite II-1 ranged from -5.9‰ to -3.6‰ (n

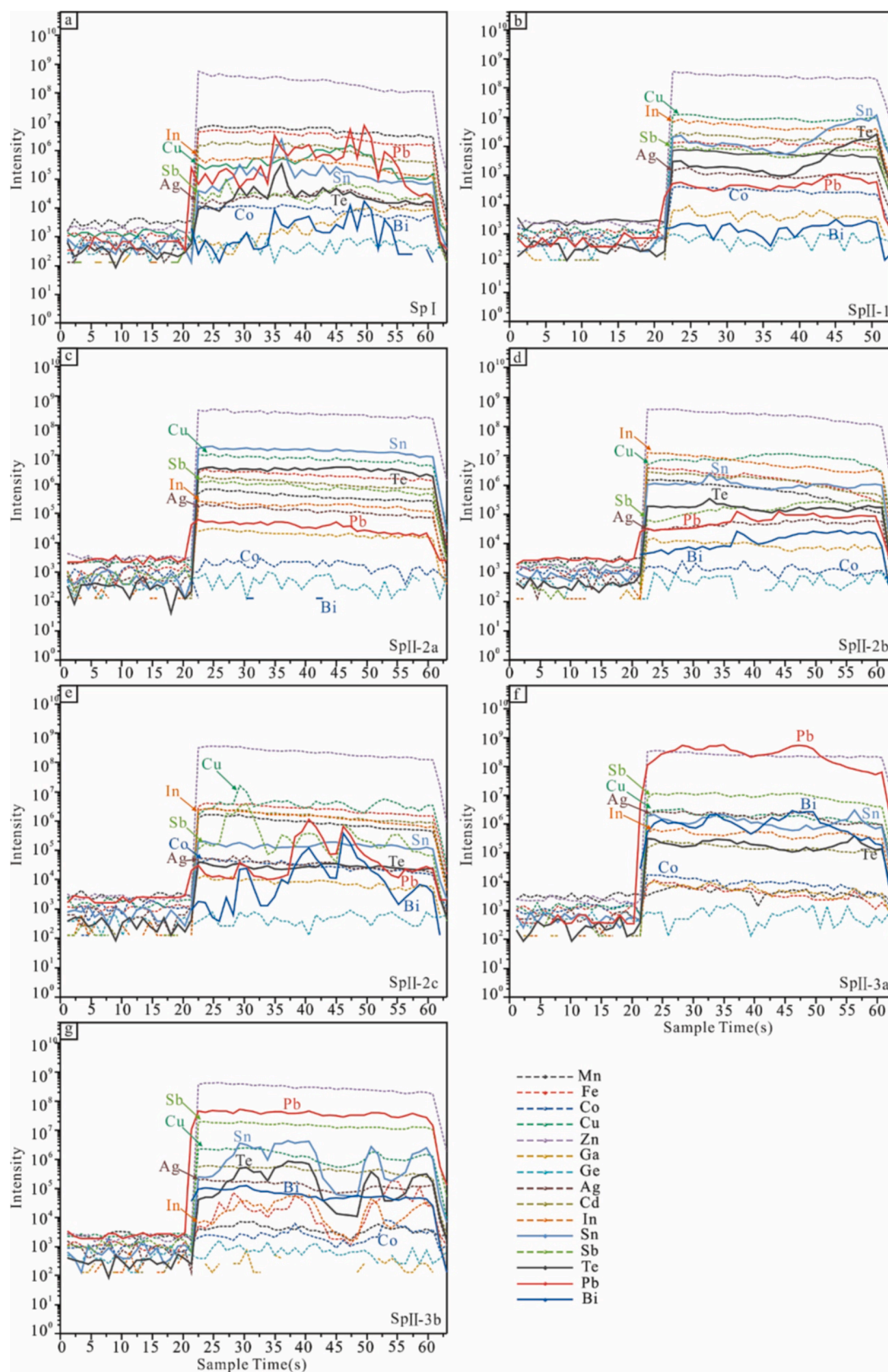


Fig. 6. Representative time-resolved LA-ICP-MS depth profiles for sphalerite in the Bayan qagan dongshan Sn-Pb-Zn-Ag deposit.

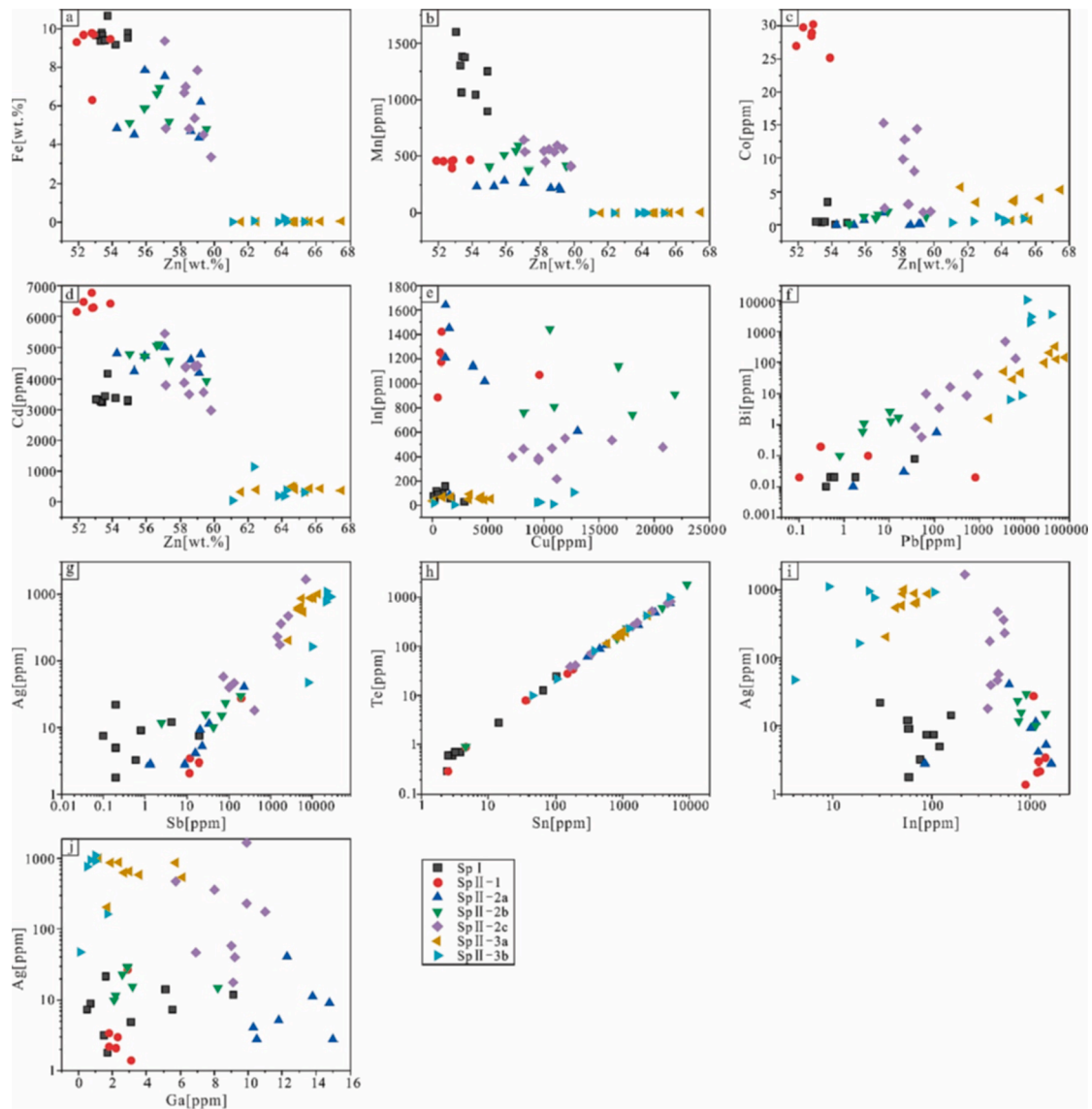


Fig. 7. Binary plots showing inter-element correlations and substitution mechanisms in sphalerite from the Bayan qagan dongshan Sn-Pb-Zn-Ag deposit. (a) Zn vs Fe; (b) Zn vs Mn; (c) Zn vs Co; (d) Zn vs Cd; (e) Cu vs In; (f) Pb vs Bi; (g) Sb vs Ag; (h) Sn vs Te; (i) In vs Ag; (j) Ga vs Ag.

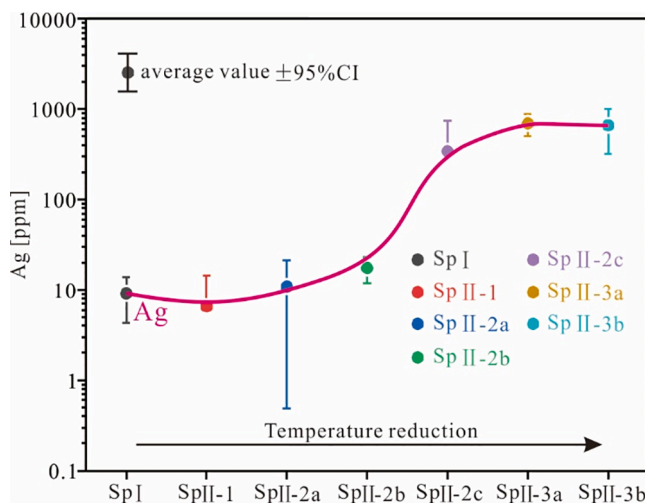


Fig. 8. Distribution of silver content in sphalerite in the Bayan qagan dongshan Sn-Pb-Zn-Ag deposit.

= 3), with an average of -4.4‰ . The $\delta^{34}\text{S}$ values for Sphalerite II-2 ranged between -4.7‰ and -1.1‰ ($n = 9$), with an average of -3.4‰ . The $\delta^{34}\text{S}$ values for Sphalerite II-3 ranged from -5.5‰ to -4.6‰ ($n = 6$), with an average of -5.1‰ .

6. Discussion

6.1. Geochemistry of trace element found in sphalerite

Previous studies have demonstrated that the crystal structure of sphalerite can accommodate a wide range of minor and trace elements, e.g., Cu, Ag, Co, Cd, Fe, Mn, Mo, Ni, Pb, Ga, Ge, In, Bi, Sn, Se and Tl (Cook et al., 2009; Ye et al., 2011), occurring mostly as solid solution with relatively stable concentrations. The variation of these elements, influenced by physicochemical parameters, are conducive to clarify the ore-forming process in detail (Cook et al., 2013; Fougereuse et al., 2016; Li et al., 2021b). However, the elements in sphalerite occur not only in the form of solid solution, but also in micro-inclusions of minerals that are of little significance to genesis of deposit. Therefore, it is necessary to identify the occurrence state and the fundamental substitution mechanisms of trace elements within sphalerite (Xing et al., 2021).

Table A1

Characteristics of the samples from the Bayan qagan dongshan Sn-Pb-Zn-Ag deposit analyzed by EPMA and LA-ICP-MS.

Sample No.	Location	Mineralization stages			Structures	Major mineral	Description
BY-B4	Granite porphyry at 860 m level	Stage I	–	–	Disseminated structure	Sphalerite	It is characterized by the disseminated sphalerite, tourmaline and a small amount of cassiterite, arsenopyrite and pyrite.
BY-B7	Granite porphyry at 860 m level	–	–	–	Disseminated structure	Sphalerite	It is characterized by the disseminated sphalerite, tourmaline and a small amount of cassiterite, arsenopyrite and pyrite.
BY-B8	III-1 orebody at 860 m level	Stage II	Stage II-1	–	Breccia structure	Cassiterite + Sphalerite + Arsenopyrite + Pyrite	Sphalerite II-1 mainly coexists with cassiterite and partially accompanies with pyrite and arsenopyrite.
BY-B9-1	III-3 orebody at 860 m level	–	Stage II-2	Stage II-2a	Massive structure + Vein structure	Sphalerite + Stannite + Galena + Pyrite	Sphalerite II-2a is surrounded by stannite and locally contains galena veinlets.
BY-B19	III-4 orebody at 860 m level	–	–	Stage II-2b	Massive structure + Vein structure	Sphalerite + Stannite + Chalcopyrite + Galena	Sphalerite II-2b is crowded with a small amount of micro inclusions of chalcopyrite, and locally crosscut by veinlets containing galena.
BY-B37	III-5 orebody at 860 m level	–	–	Stage II-2c	Massive structure + Vein structure	Sphalerite + Stannite + Chalcopyrite + Galena + Arsenopyrite	1. Sphalerite II-2c is typically characterized by the development of chalcopyrite inclusions, which is larger than that of sphalerite II-2b. 2. Sphalerite II-2c is partially cut by chalcopyrite veinlets and locally contains aggregates of fine-grained arsenopyrite.
Sample No.	Location	Mineralization stages			Structures	Major mineral	Description
BY-B9-2	III-6 orebody at 900m level	Stage II	Stage II-3	Stage II-3a	Vein structure	Sphalerite+ Galena+ Pyrite+ Stannite+ Chalcostibite	Sphalerite II-3a commonly appears as irregular intergranular aggregates and coexists with galena and minor amounts of stannite and chalcostibite.
BY-B29	III-7 orebody at 900m level	–	–	Stage II-3b	Vein structure	Sphalerite+Chalcostibite+Jamesonite+Pyrite+ Stannite	Sphalerite II-3b is intermingled with Sb-bearing minerals (e.g., tetrahedrite, chalcostibite, and Jamesonite ect.) and generally occurs as aggregates of fine-grained.

Due to the tetragonal structure of sphalerite and according to charge, radii, and coordination of ions compared with Zn^{2+} , some divalent transition metal ions including Fe^{2+} , Co^{2+} , Cd^{2+} and Mn^{2+} ($\text{Fe}^{2+} = 0.63 \text{ \AA}$, $\text{Co}^{2+} = 0.58 \text{ \AA}$, $\text{Cd}^{2+} = 0.78 \text{ \AA}$, $\text{Mn}^{2+} = 0.66 \text{ \AA}$; Shannon, 1976) are considered to be compatible with sphalerite. These elements can directly substitute for Zn^{2+} without readjustments of charge balance. The sphalerite samples from the Bayan qagan dongshan Sn-Pb-Zn-Ag deposit yield that Zn content has negative correlations with concentrations of divalent transition element, notably Fe, Cd and Mn, but also Co (Fig. 7 a-d). The good correlations between Fe, Cd and Mn and Zn suggest that Zn^{2+} can be well substituted by Fe^{2+} , Cd^{2+} and Mn^{2+} . Nevertheless, the relationship between Co and Zn shows a weak negative correlation trend, which may be mainly due to the larger Co variation in sphalerite I, sphalerite II-2a and sphalerite II-2b. The Co and Zn in the other subtypes of sphalerite show a marked negative correlation (Fig. 7c). Therefore, the Co may also enter sphalerite as a solid solution.

Copper and indium have been used to propose a $\text{Cu}^{+} + \text{In}^{3+} \leftrightarrow 2\text{Zn}^{2+}$ coupled substitution, particularly valid for the samples with the high indium contents in epithermal, skarn and stratabound deposits by Cook et al. (2009). The dataset shows that most subtype of sphalerite have a wider range of Cu variation (Table A3; Fig. 5d), which are probably caused by mineral inclusions of chalcopyrite. The result is consistent with the petrographic observations of the sphalerite II-2b and sphalerite II-2c (Fig. 3h, i). In addition, the majority of time resolved LA-ICP-MS depth for Cu in analyzed sphalerite are ragged, especially sphalerite II-2c (Fig. 6e). This result also indicates that the compositional of Cu within sphalerite grains are heterogeneity and occur predominantly as Cu-bearing mineral inclusions. In contrast, the time resolved LA-ICP-MS depth profiles of indium, except for sphalerite II-3b, are generally smooth (Fig. 6), suggesting that the indium may be homogeneity within sphalerite grains. In fact, the minority of time-resolved LA-ICP-MS depth profile characteristic for copper and indium are smooth and similar such as the curves in sphalerite II-1 and sphalerite II-2a (Fig. 6b-c). Therefore, the scattered relationship between copper and indium (Fig. 7e) was probably the influence of “chalcopyrite disease” obscuring the correlation of them.

Murakami and Ishihara (2013) proposed that silver could be incorporated into sphalerite as monovalent cations, via coupled substitutions, e.g., with Ga^{3+} , In^{3+} , Ge^{4+} , and probably also Sb^{3+} or Bi^{3+} . In this study, the time-resolved LA-ICP-MS depth profiles reveal that curves of Ag and Sb are flat and similar except for sphalerite II-2c (Fig. 6a-g), and the relationship between them also show a positive correlation (Fig. 7g), with a reasonably constant slope. Consequently, Sb and Ag may have a coupled substitution with Zn by $2\text{Zn}^{2+} \leftrightarrow \text{Ag}^{+} + \text{Sb}^{3+}$ mechanism. Although the depth profiles for indium are smooth, the scattered relationship between Ag and In (Fig. 7i) shows that the coupled substitution mechanism of them is less likely. Similarly, the curves of Ga and Ge in depth profile are flat, but Ga and Ge show no correlations with Ag (Fig. 7j). This may be due to the low content of Ga and Ge not enough to combine with Ag^{+} to substitute Zn^{2+} . For bismuth, the patterns distribution mirror those of Pb (Fig. 6a-g), both of which have raggedness characteristics, indicating that they may link to the presence of Bi- and Pb-bearing mineral inclusions. On the whole, it is considered that the composition of sphalerite in the Bayan qagan dongshan Sn-Pb-Zn-Ag deposit is at least governed by coupled substitution $2\text{Zn}^{2+} \leftrightarrow \text{Ag}^{+} + \text{Sb}^{3+}$.

Telluride, e.g., hessite (Ag_2Te), calaverite (AuTe_2), petzite (Ag_3AuTe_2), radoite (HgTe), and altaite (PbTe), is the most common tellurium-bearing minerals (Börner et al., 2021). Nevertheless, our study reveals that tellurium may combine with tin to form SnTe in the the Bayan qagan dongshan Sn-Pb-Zn-Ag deposit. In the majority of analyzed sphalerite subtypes, the time-resolved LA-ICP-MS depth profiles of Sn and Te are ragged, and they are almost identical (Fig. 6). These results, combined with positive correlation between Sn and Te as mentioned earlier, indicate that Sn and Te may be occur predominantly as SnTe inclusions within sphalerite grains.

6.2. Ore-forming temperature

It is usually considered that specific trace element compositions of sphalerite can apply to geothermometers (Cook et al., 2009; Ye et al., 2011; Frenzel et al., 2016; Bauer et al., 2019; Xing et al., 2021). Keith et al. (2014) proposed that the Fe/Zn ratio is directly proportional to

Table A2

EPMA data of sphalerite from the Bayan qagan dongshan Sn-Pb-Zn-Ag deposit (data in wt%).

Spot No.	Sphalerite subtype	Zn	As	Se	Ag	Cd	Bi	Sn	Fe	Cu	Mn	Ni	S	Te	Sb	Co	Total
BY-B4-1	Sphalerite I	53.37	0.00	0.00	0.00	0.05	0.00	0.00	11.88	0.00	0.03	0.00	34.22	0.10	0.02	0.02	99.69
BY-B4-2	Sphalerite I	53.40	0.03	0.04	0.00	0.10	0.04	0.04	11.79	0.00	0.09	0.00	34.48	0.00	0.02	0.02	100.05
BY-B4-3	Sphalerite I	53.31	0.00	0.06	0.00	0.10	0.00	0.02	11.69	0.05	0.10	0.00	33.99	0.04	0.03	0.02	99.41
BY-B4-4	Sphalerite I	53.04	0.00	0.06	0.00	0.09	0.00	0.00	11.87	0.01	0.08	0.03	34.69	0.12	0.00	0.02	100.01
BY-B4-5	Sphalerite I	53.57	0.00	0.04	0.00	0.05	0.00	0.03	11.76	0.00	0.12	0.00	33.86	0.26	0.00	0.00	99.69
BY-B4-6	Sphalerite I	54.91	0.16	0.04	0.00	0.05	0.00	0.00	11.59	0.01	0.11	0.00	33.04	0.32	0.00	0.01	100.24
BY-B4-7	Sphalerite I	54.91	0.01	0.02	0.00	0.08	0.04	0.00	11.24	0.06	0.05	0.00	33.76	0.17	0.03	0.02	100.39
BY-B4-8	Sphalerite I	54.19	0.01	0.01	0.00	0.07	0.00	0.03	11.37	0.00	0.07	0.00	34.05	0.16	0.02	0.00	99.98
BY-B7-1	Sphalerite I	53.73	0.00	0.00	0.00	0.10	0.00	0.00	11.38	0.00	0.26	0.01	34.42	0.00	0.00	0.00	99.9
BY-B8-1	Sphalerite II-1	53.88	0.00	0.00	0.00	0.13	0.00	0.29	8.02	4.40	0.06	0.00	34.73	0.07	0.00	0.03	101.61
BY-B8-2	Sphalerite II-1	52.30	0.06	0.00	0.00	0.09	0.00	0.00	11.97	0.07	0.06	0.00	34.71	0.00	0.00	0.03	99.29
BY-B8-3	Sphalerite II-1	52.78	0.05	0.00	0.00	0.06	0.00	0.00	12.08	0.08	0.02	0.00	34.46	0.31	0.00	0.02	99.86
BY-B8-4	Sphalerite II-1	52.88	0.00	0.01	0.00	0.11	0.00	0.00	12.18	0.03	0.04	0.02	34.51	0.00	0.00	0.04	99.82
BY-B8-5	Sphalerite II-1	52.80	0.07	0.00	0.00	0.16	0.00	0.00	12.15	0.01	0.04	0.00	33.40	0.00	0.00	0.02	98.65
BY-B8-6	Sphalerite II-1	51.88	0.01	0.00	0.00	0.10	0.00	0.00	11.92	0.03	0.00	0.00	34.21	0.00	0.00	0.04	98.19
BY-B9-1-2	Sphalerite II-2a	58.60	0.01	0.00	0.00	0.09	0.02	0.04	6.91	0.12	0.06	0.00	33.86	0.09	0.00	0.00	99.8
BY-B9-1-3	Sphalerite II-2a	55.92	0.00	0.04	0.00	0.09	0.00	0.02	9.49	0.03	0.04	0.02	33.95	0.00	0.00	0.00	99.6
BY-B9-1-4	Sphalerite II-2a	54.27	0.00	0.00	0.00	0.11	0.00	0.00	7.98	3.33	0.03	0.00	34.32	0.00	0.00	0.03	100.07
BY-B9-1-5	Sphalerite II-2a	57.06	0.00	0.01	0.00	0.08	0.00	0.00	7.35	0.31	0.02	0.00	34.23	0.00	0.01	0.04	99.11
BY-B9-1-6	Sphalerite II-2a	59.19	0.00	0.00	0.00	0.08	0.00	0.02	6.17	0.10	0.03	0.02	33.63	0.00	0.00	0.02	99.26
BY-B9-1-7	Sphalerite II-2a	59.09	0.00	0.00	0.00	0.09	0.01	0.00	6.35	0.13	0.03	0.00	33.72	0.15	0.00	0.02	99.59
BY-B9-1-8	Sphalerite II-2a	55.30	0.04	0.05	0	0.11	0	0	9.69	0.04	0.02	0	34.01	0	0	0.03	99.29
BY-B19-1	Sphalerite II-2b	57.32	0.00	0.00	0.00	0.05	0.02	0.00	7.47	0.32	0.07	0.02	34.30	0.21	0.00	0.03	99.81
BY-B19-2	Sphalerite II-2b	56.59	0.00	0.00	0.00	0.10	0.00	0.00	7.83	0.25	0.07	0.00	34.65	0.03	0.00	0.02	99.54
BY-B19-3	Sphalerite II-2b	55.90	0.00	0.00	0.00	0.08	0.00	0.00	8.93	0.16	0.05	0.02	34.53	0.10	0.01	0.00	99.78
BY-B19-4	Sphalerite II-2b	59.53	0.00	0.00	0.00	0.06	0.02	0.17	3.46	2.34	0.02	0.00	33.69	0.23	0.00	0.00	99.52
BY-B19-5	Sphalerite II-2b	56.73	0.00	0.00	0.00	0.04	0.00	0.00	5.98	4.75	0.06	0.01	33.83	0.00	0.01	0.00	101.41
BY-B19-6	Sphalerite II-2b	55.02	0.02	0.00	0.00	0.11	0.00	0.00	9.27	0.06	0.03	0.02	34.64	0.14	0.00	0.03	99.34
BY-B37-1	Sphalerite II-2c	57.12	0.02	0.00	0.00	0.10	0.00	0.05	7.63	0.30	0.06	0.00	34.23	0.02	0.00	0.01	99.54
BY-B37-2	Sphalerite II-2c	58.50	0.00	0.02	0.00	0.05	0.00	0.05	7.06	0.09	0.03	0.00	34.10	0.00	0.00	0.00	99.9
BY-B37-3	Sphalerite II-2c	58.21	0.00	0.00	0.00	0.09	0.00	0.00	7.02	0.22	0.05	0.01	34.03	0.00	0.00	0.01	99.64
BY-B37-4	Sphalerite II-2c	58.82	0.00	0.00	0.00	0.08	0.03	0.03	6.79	0.20	0.07	0.01	32.59	0.38	0.01	0.00	99.01
BY-B37-5	Sphalerite II-2c	58.99	0.04	0.00	0.00	0.07	0.00	0.00	6.67	0.25	0.05	0.00	34.12	0.10	0.00	0.00	100.29
BY-B37-6	Sphalerite II-2c	59.80	0.04	0.00	0.00	0.10	0.00	0.01	4.65	1.15	0.07	0.00	33.63	0.00	0.01	0.00	99.46
BY-B37-7	Sphalerite II-2c	59.35	0.00	0.01	0.00	0.02	0.00	0.00	5.67	1.13	0.05	0.00	34.18	0.00	0.00	0.01	100.42
BY-B37-8	Sphalerite II-2c	58.30	0.00	0.00	0.00	0.05	0.02	0.00	6.48	0.40	0.02	0.00	34.18	0.00	0.01	0.01	99.47
BY-B37-9	Sphalerite II-2c	57.06	0.00	0.00	0.00	0.16	0.01	0.16	5.51	3.59	0.05	0.00	33.39	0.00	0.00	0.01	99.94
BY-B9-2-1	Sphalerite II-3a	65.37	0.00	0.03	0.00	0.03	0.00	0.00	0.03	0.96	0.00	0.00	33.60	0.00	0.95	0.00	100.97
BY-B9-2-2	Sphalerite II-3a	61.57	0.03	0.00	0.01	0.02	0.00	0.00	0.04	2.05	0.00	0.04	33.27	0.00	2.46	0.01	99.5
BY-B9-2-3	Sphalerite II-3a	62.49	0.00	0.00	0.00	0.00	0.00	0.00	0.18	1.94	0.03	0.00	33.21	0.00	2.64	0.01	100.5
BY-B9-2-4	Sphalerite II-3a	65.64	0.00	0.00	0.00	0.00	0.01	0.00	0.06	0.48	0.00	0.00	33.36	0.00	1.04	0.00	100.59
BY-B9-2-5	Sphalerite II-3a	64.77	0.00	0.02	0.00	0.00	0.00	0.00	0.07	0.41	0.00	0.00	33.61	0.00	0.76	0.00	99.64
BY-B9-2-6	Sphalerite II-3a	67.48	0.02	0.00	0.00	0.01	0.00	0.01	0.12	0.13	0.01	0.01	33.40	0.00	0.20	0.01	101.4
BY-B9-2-7	Sphalerite II-3a	66.26	0.00	0.01	0.00	0.04	0.00	0.01	0.04	0.07	0.00	0.00	33.35	0.00	0.36	0.00	100.14
BY-B9-2-8	Sphalerite II-3a	64.55	0.00	0.00	0.00	0.01	0.02	0.01	0.10	0.21	0.01	0.00	33.79	0.00	0.58	0.00	99.28
BY-B9-2-9	Sphalerite II-3a	64.69	0.00	0.00	0.00	0.03	0.00	0.02	0.00	1.15	0.01	0.01	33.49	0.00	1.07	0.00	100.48
BY-B29-1	Sphalerite II-3b	63.77	0.09	0.00	0.00	0.01	0.00	0.00	0.07	0.68	0.00	0.01	33.55	0.00	1.36	0.01	99.55
BY-B29-2	Sphalerite II-3b	65.30	0.00	0.00	0.00	0.01	0.06	0.00	0.06	0.09	0.02	0.04	32.98	0.00	0.95	0.01	99.52
BY-B29-3	Sphalerite II-3b	64.14	0.04	0.00	0.00	0.00	0.00	0.00	0.02	0.18	0.02	0.01	33.02	0.00	1.45	0.01	98.89
BY-B29-4	Sphalerite II-3b	64.27	0.06	0.00	0.00	0.05	0.00	0.00	0.07	0.28	0.00	0.00	33.73	0.00	1.58	0.00	100.04
BY-B29-5	Sphalerite II-3b	61.06	0.00	0.00	0.00	0.00	0.00	0.00	0.24	1.73	0.00	0.02	32.90	0.00	3.30	0.00	99.25
BY-B29-6	Sphalerite II-3b	62.33	0.00	0.00	0.00	0.00	0.00	0.00	0.13	1.06	0.02	0.01	33.22	0.00	2.27	0.00	99.04

Note: The value of '0.00' means the content of element are approximately zero.

formation temperature of sphalerite. Möller (1987) believe that the Ga/Ge ratio of sphalerite can serve as formation temperature detection. Zhou et al. (2009) and Zhou et al. (2012) suggest that Zn/Cd, Ga/In, and Ge/In ratios were related to the crystallization temperature of sphalerite. However, none of the above methods can accurately estimate the sphalerite formation temperature. For example, Ga/Ge ratio as a geological thermometer parameter is only effective for Pb-Zn deposits below 300 °C, and Fe/Zn, Zn/Cd, Ga/In, and Ge/In ratio can not quantitatively reflect sphalerite formation temperature.

New research shows that the contents of Fe, Mn, Ga, Ge, and In within sphalerite are intimately related to the crystallization temperature of sphalerite (Frenzel et al., 2016). According to research, the empirical relationship between trace elements and the homogenization

temperature, T, is: $T(^{\circ}\text{C}) = -(54.4 \pm 7.3) \times \text{PC1}^* + (208 \pm 10)$, where PC1* expressed as $\ln[(\text{C0.22 Ga} \times \text{C0.22 Ge})/(\text{C0.37 Fe} \times \text{C0.20 Mn} \times \text{C0.11 In})]$ with Ga, Ge, In and Mn concentrations in ppm, and Fe concentration in wt.%. The formation temperatures of different sphalerite subtypes were calculated as follows: the temperature of sphalerite I is between 325.1 and 384.5 °C, with an average of 354.8 °C; the temperature of sphalerite II-1 is between 297.5 and 348.4 °C, with an average of 322.9 °C; the temperature of sphalerite II-3 is between 146.4 and 179.7 °C, with an average of 163.0 °C. It is essentially consistent with the homogenization temperature of fluid inclusions, which is between 355 °C and 129 °C (Unpublished data). The results show that the ore-forming temperature gradually decreased from early to late stage. Furthermore, the results also indicate that the sphalerite was formed in a

Table A3

LA-ICP-MS data of sphalerite from the Bayan qagan dongshan Sn-Pb-Zn-Ag deposit (data in ppm).

Spot No.	Sphalerite subtype	⁵⁵ Mn	⁵⁹ Co	⁶⁵ Cu	⁷¹ Ga	⁷² Ge	¹⁰⁷ Ag	¹¹¹ Cd	¹¹⁵ In	¹¹⁸ Sn	¹²¹ Sb	⁸² Se	¹²² Te	²⁰⁸ Pb	²⁰⁹ Bi
BY-B4-L_1	Sphalerite I	1064.0	0.39	881.0	5.5	1.96	7.4	3328.0	104.8	3.2	0.1	106.70	0.7	0.6	<0.01
BY-B4-L_2	Sphalerite I	1383.4	0.47	387.0	3.1	1.38	4.9	3254.0	120.1	2.9	0.2	152.84	0.6	0.8	<0.00
BY-B4-L_3	Sphalerite I	1303.3	0.48	1142.0	5.1	1.63	14.3	3279.0	157.0	2.4	<0.05	157.66	0.3	0.5	0.02
BY-B4-L_4	Sphalerite I	1601.9	0.44	2884.0	1.6	0.56	21.9	3337.0	30.2	14.4	0.2	26.01	2.8	1.1	<0.00
BY-B4-L_5	Sphalerite I	1376.7	0.44	170.0	1.7	<0.39	1.8	3441.0	59.2	3.8	0.2	<9.36	0.7	0.6	0.02
BY-B4-L_6	Sphalerite I	1252.1	0.32	84.0	1.5	<0.27	3.2	3281.0	75.8	2.5	0.6	23.70	0.6	1.8	0.02
BY-B4-L_7	Sphalerite I	896.4	<0	1618.0	0.7	0.6	9.0	3312.0	58.3	3.2	0.8	26.18	0.7	0.4	0.01
BY-B4-L_8	Sphalerite I	1044.2	0.05	1618.0	9.1	<0.26	12.0	3383.0	57.4	64.6	4.2	<0.00	12.7	1.4	<0.00
BY-B7-L_1	Sphalerite I	3021.5	3.46	469.0	0.5	0.13	7.4	4175.0	88.3	102.4	19.4	9.26	24.3	36.2	0.08
BY-B8-L_1	Sphalerite II-1	466.1	25.2	956.6	2.3	<0.30	3.0	6,423.8	1,216.7	148.9	18.8	12.09	28.1	0.3	0.2
BY-B8-L_2	Sphalerite II-1	454.6	29.8	800.3	1.8	<0.39	3.4	6,481.7	1,422.9	36.1	11.4	21.10	7.9	0.2	<0.00
BY-B8-L_3	Sphalerite II-1	454.4	28.5	669.0	1.8	<0.32	2.2	6,777.1	1,253.3	4.6	<0.08	12.00	0.9	<0.00	<0.00
BY-B8-L_4	Sphalerite II-1	465.0	30.3	456.6	3.1	<0.31	1.4	6,295.2	889.4	2.5	<0.00	14.42	0.3	0.1	0.02
BY-B8-L_5	Sphalerite II-1	397.4	29.0	9,580.9	2.9	0.6	27.4	6,281.2	1,075.7	326.7	195.8	25.79	68.3	3.4	0.1
BY-B8-L_6	Sphalerite II-1	458.7	27.0	767.2	2.2	<0.36	2.1	6,161.1	1,175.3	180.4	11.1	<5.65	34.4	802.0	0.02
BY-B9-L_1-L_2	Sphalerite II-2a	221.4	0.0	1,515.2	11.8	<0.21	5.2	4,595.2	1,450.7	849.0	22.7	<2.40	157.7	21.6	0.03
BY-B9-L_1-L_3	Sphalerite II-2a	286.4	0.7	1,565.8	15.0	1.1	2.8	4,725.0	85.1	453.4	1.3	<0.00	87.8	1.7	<0.00
BY-B9-L_1-L_4	Sphalerite II-2a	236.8	0.0	3,666.8	13.8	<0.00	11.3	4,812.3	1,142.2	3,075.6	33.7	<6.24	485.9	5.4	<0.00
BY-B9-L_1-L_5	Sphalerite II-2a	266.3	1.9	1,164.6	10.3	0.6	4.1	4,996.1	1,210.2	566.0	15.6	<4.17	106.3	18.7	<0.00
BY-B9-L_1-L_6	Sphalerite II-2a	204.1	0.1	4,735.8	14.8	<0.00	9.2	4,780.4	1,020.4	5,141.2	20.3	<1.31	761.1	6.9	<0.00
BY-B9-L_1-L_7	Sphalerite II-2a	227.2	0.1	1,193.2	10.5	<0.40	2.8	4,196.7	1,641.5	300.0	8.7	<1.67	60.7	1.6	0.01
BY-B9-L_1-L_8	Sphalerite II-2a	237.6	0.0	13,077.2	12.3	0.1	40.5	4,245.2	611.3	1732.0	231.5	<3.72	268.2	111.4	0.57
BY-B19-L_1	Sphalerite II-2b	379.9	2.0	18,027.0	2.6	<0.12	23.2	4,561.6	745.8	1,165.1	82.8	<6.04	231.2	16.3	1.7
BY-B19-L_2	Sphalerite II-2b	546.1	1.0	16,762.5	2.1	<0.19	10.0	5,053.0	1,138.6	3,922.7	42.2	<6.22	599.0	2.6	0.6
BY-B19-L_3	Sphalerite II-2b	509.4	1.2	10,939.7	3.2	<0.24	15.6	4,701.7	810.4	817.0	27.7	9.80	153.0	2.8	1.1
BY-B19-L_4	Sphalerite II-2b	415.8	1.2	21,867.5	2.9	0.7	29.2	3,930.1	913.5	9,186.5	192.8	<6.96	1,808.9	10.6	2.7
BY-B19-L_5	Sphalerite II-2b	591.6	1.5	8,267.0	2.2	<0.02	11.6	5,094.7	763.1	4.6	2.4	10.39	0.9	0.8	0.1
BY-B19-L_6	Sphalerite II-2b	407.4	0.1	10,553.0	8.2	0.0	14.9	4779.3	1441.1	828.8	67.2	<3.95	142.5	11.1	1.30
BY-B37-L_1	Sphalerite II-2c	536.4	2.5	10,738.5	5.7	<0.00	474.3	3,792.5	467.7	1,102.8	2,658.0	<5.91	210.0	6,351.8	131.0
BY-B37-L_2	Sphalerite II-2c	559.3	3.1	9,517.5	11.0	0.6	174.9	3,500.2	388.2	4,717.4	1,678.3	<4.70	748.8	917.6	42.9
BY-B37-L_3	Sphalerite II-2c	543.8	9.8	7,228.1	9.2	<0.31	39.6	3,870.1	397.4	1,409.8	101.1	<4.10	253.1	126.3	3.5
BY-B37-L_4	Sphalerite II-2c	533.8	8.1	11,182.3	9.9	0.7	1,682.3	4,400.5	216.8	340.9	7,006.0	<8.24	70.1	3,732.5	468.5
BY-B37-L_5	Sphalerite II-2c	594.1	14.4	8,242.8	6.9	<0.42	46.5	4,411.9	464.0	197.0	133.9	<0.00	41.1	51.1	0.4
BY-B37-L_6	Sphalerite II-2c	412.0	2.0	16,162.3	8.0	2.3	360.2	2,978.0	535.1	5,213.9	1,785.1	<0.00	810.4	222.3	16.7
BY-B37-L_7	Sphalerite II-2c	562.9	1.9	11,930.6	9.9	<0.00	230.7	3,566.6	550.1	2,758.1	1,447.9	<5.14	515.5	521.9	8.9
BY-B37-L_8	Sphalerite II-2c	451.2	12.8	20,781.8	9.0	3.7	57.6	4,368.8	477.5	1,664.5	73.4	<5.68	303.6	36.9	0.8

(continued on next page)

Table A3 (continued)

Spot No.	Sphalerite subtype	⁵⁵ Mn	⁵⁹ Co	⁶⁵ Cu	⁷¹ Ga	⁷² Ge	¹⁰⁷ Ag	¹¹¹ Cd	¹¹⁵ In	¹¹⁸ Sn	¹²¹ Sb	⁸² Se	¹²² Te	²⁰⁸ Pb	²⁰⁹ Bi
BY-B37-L_9	Sphalerite II-2c	641.6	15.3	9,528.4	9.1	0.2	17.9	5442.3	370.2	163.4	411.2	<5.96	38.1	64.4	10.07
BY-B9-2-L_1	Sphalerite II-3a	4.8	1.2	1,689.1	2.4	0.8	884.5	344.9	66.7	820.8	8,975.0	50.97	163.7	48,200.0	322.2
BY-B9-2-L_2	Sphalerite II-3a	0.5	5.7	4,701.8	6.1	0.7	544.6	330.6	43.9	976.5	6,174.4	<6.89	197.2	8,283.2	47.4
BY-B9-2-L_3	Sphalerite II-3a	1.3	3.4	3,281.3	3.6	0.2	588.2	398.9	50.2	819.8	4,723.5	22.87	152.5	5,506.2	29.6
BY-B9-2-L_4	Sphalerite II-3a	1.1	0.7	883.1	2.7	0.5	631.3	440.4	67.2	1,076.4	4,658.0	21.48	201.2	36,148.6	201.4
BY-B9-2-L_5	Sphalerite II-3a	1.0	3.8	4,440.2	3.0	0.4	656.7	412.8	70.2	2,382.8	5,478.8	21.30	418.9	3,469.2	53.2
BY-B9-2-L_6	Sphalerite II-3a	7.4	5.3	5,298.9	1.2	0.3	1,014.6	377.1	53.0	902.6	13,727.1	23.58	170.7	52,379.3	123.7
BY-B9-2-L_7	Sphalerite II-3a	6.6	4.0	4,322.0	1.9	0.2	873.1	437.2	51.7	1,125.4	11,008.4	25.93	191.2	28,030.8	95.9
BY-B9-2-L_8	Sphalerite II-3a	0.5	0.6	45.2	1.7	0.1	203.5	498.2	34.9	589.3	2,721.4	11.45	114.7	1,618.9	1.6
BY-B9-2-L_9	Sphalerite II-3a	1.0	3.5	3,355.3	5.7	0.1	871.3	490.8	90.9	967.2	5601.6	34.25	158.2	80162.9	143.19
Spot No.	Sphalerite subtype	⁵⁵ Mn	⁵⁹ Co	⁶⁵ Cu	⁷¹ Ga	⁷² Ge	¹⁰⁷ Ag	¹¹¹ Cd	¹¹⁵ In	¹¹⁸ Sn	¹²¹ Sb	⁸² Se	¹²² Te	²⁰⁸ Pb	²⁰⁹ Bi
BY-B29-L_1	Sphalerite II-3b	0.2	1.2	10,844.7	1.0	0.1	1,110.8	202.4	9.1	103.1	22,237.3	<7.43	21.4	13,997.8	2,935.3
BY-B29-L_2	Sphalerite II-3b	1.5	0.9	9,389.6	0.7	0.4	955.9	313.5	23.3	375.5	22,190.4	<3.59	80.5	39,790.9	3,541.1
BY-B29-L_3	Sphalerite II-3b	5.9	0.5	12,689.4	1.0	0.2	925.8	195.5	106.1	5,066.5	27,911.7	<9.03	994.7	11,441.2	10,435.5
BY-B29-L_4	Sphalerite II-3b	0.6	0.6	9,636.5	0.5	0.4	770.4	383.1	26.4	2,262.6	21,572.5	<1.05	418.1	13,633.7	1,951.3
BY-B29-L_5	Sphalerite II-3b	2.4	0.3	106.8	1.7	0.0	163.4	45.3	18.5	45.0	9948.0	<5.82	9.9	4772.9	6.52
BY-B29-L_6	Sphalerite II-3b	0.7	0.5	1,961.7	0.1	0.4	47.1	1140.0	4.1	1241.0	8082.5	18.02	229.2	8786.8	8.99

Note: The data denoted with '<' are below the minimum detection limits.

Table A4

LA-MC-ICP-MS in situ S isotope compositions of various sphalerite subtypes from the Bayan qagan dongshan Sn-Pb-Zn-Ag deposit.

Analysis spot	Sphalerite subtype	$\delta^{34}\text{S}_{\text{V-CDT}}(\text{‰})$	Error (2 σ)
BY-B4-S1	Sphalerite I	-6.0	0.2
BY-B4-S2	Sphalerite I	-5.8	0.3
BY-B4-S3	Sphalerite I	-3.6	0.3
BY-B8-S4	Sphalerite II-1	-5.9	0.1
BY-B8-S5	Sphalerite II-1	-3.6	0.2
BY-B8-S6	Sphalerite II-1	-3.8	0.1
BY-B9-S7	Sphalerite II-2a	-3.7	0.2
BY-B9-S8	Sphalerite II-2a	-4.0	0.2
BY-B9-S9	Sphalerite II-2a	-4.6	0.2
BY-B19-S11	Sphalerite II-2b	-4.3	0.3
BY-B19-S12	Sphalerite II-2b	-4.3	0.2
BY-B19-S13	Sphalerite II-2b	-4.7	0.2
BY-B37-S14	Sphalerite II-2c	-2.5	0.7
BY-B37-S15	Sphalerite II-2c	-1.7	0.7
BY-B37-S16	Sphalerite II-2c	-1.1	0.9
BY-B9-S19	Sphalerite II-3a	-4.7	0.1
BY-B9-S20	Sphalerite II-3a	-5.2	0.2
BY-B9-S21	Sphalerite II-3a	-5.5	0.1
BY-B29-S23	Sphalerite II-3b	-5.3	0.3
BY-B29-S28	Sphalerite II-3b	-4.6	0.2
BY-B29-S29	Sphalerite II-3b	-4.6	0.2

high-temperature environment in the early and middle stage (e.g., sphalerite I, sphalerite II-1, and sphalerite II-2), while sphalerite was formed in a low-temperature environment in the late stage (e.g., sphalerite II-3).

6.3. The mineralization process

According to relevant researches, the silver in the ore of the Bayan qagan dongshan Sn-Pb-Zn-Ag deposit mainly hosted in the tetrahedrite (90.06 %), sphalerite (3.33 %), galena-jamesonite (2.86 %), pyrite-arsenopyrite (1.78 %) (Liang et al., 2020). The results show silver in Bayan qagan dongshan Sn-Pb-Zn-Ag deposit is mostly precipitated in S-bearing environment. Previous studies have demonstrated that AgHS and Ag(HS)-2 are the main forms for silver in S-bearing hydrothermal fluid with decreasing temperature and salinity (Seward, 1976; Gammons and Barnes, 1989; Stefansson and Seward, 2003), and Ag deposition are mainly associated with physical conditions changing (e.g., $f\text{O}_2$, pH and pressure) and fluid mixing, as well as fluid-rock interaction (Spycher and Reed, 1989; Seward, 1989; Berger et al., 2003; Ronacher et al., 2004; Simmons et al., 2005; Zhai et al., 2018). Seven sphalerite subtypes observed in the Bayan qagan dongshan deposit provide a vital clue to link distinct mechanisms with Ag precipitation.

As mentioned earlier, the ore forming temperatures of Bayan qagan dongshan Sn-Pb-Zn-Ag deposit gradually decreased. Nevertheless, conductive cooling has insignificant effects on migration and precipitation of silver, because the solubility of silver sulphide with regard to AgHS is almost independent of temperature, and Ag(HS)-2 solubility changes minor under the condition of slow cooling (Stefansson and Seward, 2003). In fact, a significant decrease of temperature and sulfur fugacity may be trigger precipitation of submicroscopic precious metals. Rapid growth rate (e.g., fluid mixing and cooling) may cause the formation of intrinsic defects or interfaces in sphalerite, resulting in the incorporation of monovalent Ag during sphalerite crystallization (Wu et al., 2018). It is evident that textures observed in Ag-rich sphalerite subtypes (e.g., Sp II-3a and Sp II-3b) are commonly irregular shapes

(Fig. 3j-k), which is most likely caused by cooling crystallization rapidly. However, Ag-poor sphalerite subtypes (e.g., Sp I, Sp II-1, Sp II-2a, Sp II-2b and Sp II-2c) generally have coarse-crystalline, subhedral to euhedral characteristics (Fig. 3b, d, g-i), suggesting that their growth rate relatively slower without rapid temperature change.

Actually, the higher concentrations of invisible Ag in Ag-rich sphalerite subtypes relative to Ag-poor equivalents may be mainly caused by the change of oxidation in hydrothermal fluid, which is similar to the paradigm that Gammons and Barnes (1989) mentioned. The dataset shows the anion of Ag-rich sphalerite subtypes coexisting with Sb-bearing tetrahedrite and jamesonite, e.g., SpII-3b, is S^{2-} , SbS_3^{2-} and $Sb_3S_7^{5-}$, whereas that of Ag-poor sphalerite subtypes, pyrite, chalcopyrite assemblage (e.g., SpII-1, SpII-2a, SpII-2b and SpII-2c) is only S^{2-} . This result indicates that silver is more easily precipitated in response a increase in reduced sulfur, or in other word oxidation state decrease (e.g., apart from Ag-rich sphalerite, the content of silver in Ag-Sb bearing tetrahedrite is up to 6.4 %). Notably, the increase of Sb content with the decrease of temperature may be another reason (Fig. 5l), which effectively promote the occurrence of $2Zn^{2+} \leftrightarrow Ag^+ + Sb^{3+}$ coupled substitution mechanism, forming Ag-Sb-bearing sphalerite.

Stefansson and Seward (2003), have summarized the solubility of silver sulphide data and demonstrated that silver sulphide saturation is affected by degassing of H_2S and pH increase after boiling process (e.g., $Ag_2S(s) + H_2S(aq) = 2AgHS(aq)$ and $Ag_2S(s) + H_2S(aq) + 2HS^- = 2Ag(HS)_2^-$). Within the Bayan qagan dongshan deposit, more than 4 types of breccia, accompanied by a large amount of tourmaline and fluorite, are known (Yao et al., 2021). Furthermore, the coexistence of vapor-rich and liquid-rich inclusions were found in the cassiterite-sulfide stage (Unpublished data). These allow us to conclude that fluid occurs boiling during the transport of silver complexes. The boiling also supported by the fact that the $\delta^{34}S$ value of the deposit display a large variation. The measured $\delta^{34}S$ values on sphalerite I (-6.0 to -3.6 ‰), sphalerite II-1 (-5.9 to -3.8 ‰), sphalerite II-2 (-4.7 to -1.1 ‰) and sphalerite II-3 (-5.5 to -4.6 ‰), combined with existing test result (the $\delta^{34}S$ values of spherite, which comes from granite porphyry and is equivalent to Sp I, range from -10.6 to -9.0 ‰; Yao et al., 2021), show that the $\delta^{34}S$ values of different subtypes sphalerite range from -10.6 ‰ to -1.1 ‰. Several studies have suggested that the sulfur isotope composition of a mineral is decided, apart from sulfur source, by the physicochemical conditions of the system (Ohmoto, 1986; Hutchison et al., 2020; Li et al., 2021b), especially the pressure change, which is considered to be a driving force for fluid flow associated vein systems (Cox et al., 2001; Weatherley and Henley, 2013). During the whole ore-forming process, the reduction of pressure caused loss of H_2S and decrease of H_2S/SO_4^{2-} , resulting in phase separation and the enrichment of ^{34}S isotopes in residual ore fluid, forming a relatively wide range in $\delta^{34}S$ (Drummond and Ohmoto, 1985; Durring et al., 2009; Kamvong and Zaw, 2009; Hodkiewicz et al., 2009; Shan, et al., 2009; Hutchison et al., 2020).

In general, changes in physicochemical conditions, especially temperature and pressure, may be the most important mechanisms of Ag precipitation in the deposit. Pressure drop and rapid cooling (e.g., SpII-2b to SpII-3a stage; Fig. 8) effects on ore-forming fluids migration processes probably play an important role.

6.4. The mineralization type

The trace element signatures of sphalerite are considered to be intimately related to the genetic types of deposits (Cook et al., 2009; Belissont et al., 2014). Cook et al. (2009) demonstrated that Ga and Ge contents of sphalerite are low in deposits related to magmatic or volcanic activities, while indium is chiefly enriched in sphalerite of hydrothermal ore deposits. Zhou et al. (1989) and Han (1994) proposed that Ga/In and Ge/In ratio in sphalerite are <1 and 0.1 in hydrothermal ore deposits, respectively. The sphalerite from Bayan qagan dongshan Sn-Pb-Zn-Ag deposit is a poor host for both Ga and Ge and a rich host for

In (Table A3), leading to Ga/In and Ge/In ratio <1 and 0.1, respectively. The results show that the genetic type of Bayan qagan dongshan Sn-Pb-Zn-Ag deposit may belong to hydrothermal ore deposit. In addition, Yao et al. (2021) found that the lead isotope composition of Bayan qagan dongshan granite porphyry is essentially consistent with that of sphalerite and galena in the deposits, indicating that the granite porphyry may be the main material source of Pb in the deposit. Furthermore, available ages for mineralization and magmatism also support a hydrothermal origin for the deposit surrounding the Bayan qagan dongshan granite porphyry (Fig. 2a), such as the measured U-Pb cassiterite age from vein orebody (140 ± 12 Ma; Unpublished data) is comparable to the zircon U-Pb ages for the Bayan qagan dongshan granite porphyry (141.7 ± 0.8 Ma and 140.2 ± 1.1 Ma, Yao et al., 2017; 141.4 ± 4.6 Ma, Liu et al., 2017b). The latest research shows that the genetic type of tin or silver polymetallic deposits in the SGKR is mainly hydrothermal vein deposits related to granitic rocks (Ouyang et al., 2015; Zhu et al., 2016; Liu et al., 2018b). The Bayan qagan dongshan Sn-Pb-Zn-Ag deposit has neither typical alteration zoning of porphyry type deposit nor mineral characteristics of greisen type tin polymetallic deposit. Combined with regional data, the deposit is likely to be a hydrothermal vein type Sn-Pb-Zn-Ag polymetallic deposit controlled by fault structures.

During the ore-forming process of Bayan qagan dongshan Sn-Pb-Zn-Ag deposit, it produce the distinct type of ores with different mineral assemblages, textures, and sulfur isotope compositions. Within the Bayan qagan dongshan deposit, tin mainly occurs as cassiterite and stannite, and may be present as tellurium complexes in ore-forming fluid at medium-low temperature, because the Te/Sn ratios measured by LA-ICP-MS in sphalerites and stannite were constant at 1:5. Silver mainly occurs in tetrahedrite and sphalerite, and may be present as $AgHS$ and $Ag(HS)_2^-$ complexes in ore-forming fluid. Its precipitation is mostly controlled by the temperature and pressure change. In addition, a synthesis of published data indicate that the deposits characterized by breccia and vein with a complex paragenesis of oxides and sulphides are important Sn-Ag mining industry type. These deposits are mainly distributed in Bolivia and the Russian Far East, of which Cerro Rico is the largest silver deposit in the world. It contains proven reserves of ~84000 tons Ag and ~100 million tons Sn (Grant et al., 1980; Bartos, 2000). According to relevant research, this type of Sn-Ag deposit is not only related to magmatism, but also related to contemporaneous or late volcanic activity. At present, no volcanic rocks contemporaneous with granite porphyry have been found in the Bayan qagan dongshan Sn-Pb-Zn-Ag deposit. Therefore, the periphery of deposit still has a good prospecting.

7. Conclusion

(1) Seven subtypes of sphalerite (SpI, SpII-1, SpII-2a, SpII-2b, SpII-2c, SpII-3a and SpII-3b) were identified according to mineral assemblages in the quartz-tourmaline-sericite-sphalerite stage (I) and cassiterite-sulfide stage (II) of the Bayan qagan dongshan Sn-Pb-Zn-Ag deposit. Compared with the late stage sphalerite subtype, the early stage sphalerite is relatively rich in Fe, Cd, Mn, Co and In, but poor in Zn, Ag, Bi, Pb, Sb and Sn.

(2) Representative time-resolved depth profile and inter-element correlation analyses based on the LA-ICP-MS method illustrated that: 1) Fe, Cd, Mn, Co, Cu, In, Ag and Sb are present in solid solutions formed through direct cation substitution as $Zn^{2+} \leftrightarrow (Fe^{2+}, Cd^{2+}, Mn^{2+}, Co^{2+})$ and coupled substitution as $Zn^{2+} \leftrightarrow Cu^{+} + In^{3+}$ and $Zn^{2+} \leftrightarrow Ag^{+} + Sb^{3+}$; 2) Pb, Bi, Sn and Te may be closely related to Sn-bearing telluride and galena inclusions hosted by sphalerite.

(3) Sphalerite geothermometer research indicate that the ore-forming temperature of the Bayan qagan dongshan Sn-Pb-Zn-Ag deposit gradually decreases and the silver mineralization mainly occurred in the late stage.

(4) Sphalerite textures, trace element compositions and sulfur isotope compositions suggest that silver is mainly precipitated in

response to changes of physicochemical conditions, such as temperature and pressure.

(5) Sphalerite trace element compositions, combined to chronological evidence, show that the Bayan qagan dongshan deposit may belong to magmatic-hydrothermal origin.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A

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